The Cosmological Constant as a Contingent Modulus: A Six Birds Demarcation of Dark Energy

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Abstract

Dark energy poses several different questions at once: what kind of term the cosmological constant is, what fixes its value, why its density is comparable to that of matter today, and what current surveys say about an evolving equation of state. We study these questions in finite-dimensional mathematical models built with the Six Birds closure calculus, and for each model we separate what it determines from what it leaves as input. First, in a convex closure problem that attains its optimum and whose optimal value is finite and differentiable near the budget level, a budget whose relaxation strictly lowers the optimal cost carries a single positive price; in an allocation model with an originally binding budget, a common linear utility term leaves the optimizer unchanged and shifts only that price, as long as the shifted price stays nonnegative. Second, in a self-consistent lattice model, the convergence readouts used to admit background values do not determine the background: all eighteen sampled offsets converge, fifteen pass the sixty-update test, and neither readout recovers the input. When a global density ledger is added, the absolute background becomes a gauge coordinate that the invariant record cannot recover, and no translation-invariant probability distribution over it exists. Third, in a toy universe with structure growth and record erasure, at least ten of twelve sampled populations retain distinguishable records exactly for $-1/8000 < \lambda \leq \lambda_+$, with $3 < \lambda_+ < 3.4$, but the ratio $|\lambda|/\rho_m$ at the first record is not pinned near unity. Whether the present epoch is typical then depends on a declared prior: none of five tested priors meets a declared concentration criterion, and smoothness near zero is not enough. On DESI DR2, Pantheon+ and compressed Planck data, pre-registered audits find a fitted rewrite shape compatible with a constant, show that the best-fit evolving history has a positive, regular density through its $w = -1$ crossing, and flag structured dependence between supernova and BAO subsets, led by high-redshift supernovae. The statements labelled as Lean theorems are kernel-checked. None of these results identifies a model price with the physical cosmological constant or derives its value.

Keywords. dark energy; cosmological constant; shadow prices; identifiability; coincidence problem; Lean; emergence calculus; Six Birds Theory.

1 Introduction

1.1 Several questions under one name

Observationally, dark energy enters cosmology as one fitted number, $\Omega_\Lambda \approx 0.7$ [26, 24]. Around that number sits a cluster of separate puzzles [1]:

1. the **cosmological-constant problem**: naive quantum-field-theoretic estimates of the vacuum energy exceed the observed value by up to about 120 orders of magnitude [36, 23];
2. the **Hubble tension**: early- and late-universe routes to H_0 disagree at the several-σ level [12];

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3. **dataset inconsistencies**: supernova and BAO distances need not agree, and comparing standard candles with standard rulers gives a distance-duality consistency test [2];
4. **phantom crossing**: some joint fits of DESI DR2 with CMB and supernova data favor evolving equation-of-state histories that cross $w = -1$ [11], whereas a canonical, minimally coupled scalar field with positive energy density cannot cross, and scalar theories whose Lagrangian depends only on the field and its first derivatives cannot cross stably under regularity assumptions [5, 34], although some higher-derivative theories can [10];
5. the **growth or S_8 tension**: some low-redshift structure probes sit mildly below CMB-based expectations [17];
6. the **coincidence problem**: why the dark-energy and matter densities are comparable now [33];
7. **interacting dark-sector models** and
8. **modified-gravity models**, both of which remain hard to separate from a cosmological constant on current data [7, 18];
9. the **anthropic question**: whether selection arguments should replace a derivation [35].

These items are usually treated as one problem, to be solved by deriving the value of Λ or predicting $w(z)$ [8]. They are in fact different kinds of question. Some ask what *kind* of term the constant is. One asks what fixes its *value*. Some ask which values are compatible with a universe that forms structure. One asks whether our epoch is *typical*. And several are questions about the *data*: which fits are preferred, and whether different probes agree.

1.2 Approach

This paper treats these questions separately, using the closure calculus of the Six Birds program [32, 31] as a common vocabulary. In that calculus a theory is a *closure package*: a coarse description, a rule for filling in what the description leaves out, and audits that test the result (Section 2). Two of its notions do most of the work here. A **budget** with its **shadow price** describes a global constraint and the multiplier that enforces it. A **readout** is what a description records; asking whether one readout determines a quantity is a precise question with a yes-or-no answer.

For the questions studied here we specify finite-dimensional models or data audits, state exactly what they contain, and ask what they determine. The answers take three forms: theorems about the model (most of them kernel-checked in Lean 4), computations on the model with controls that could have failed, and pre-registered audits of public data. We call the resulting separation of determined from undetermined parts a *demarcation*. It is always relative to a stated model and a stated readout.

1.3 Results in brief

1. **The constant slot as a price (Section 4).** If a convex closure problem attains its optimum and its optimal value is finite near the budget level and strictly improves when the budget is relaxed, the budget binds and carries a positive scalar price, with full Lagrangian optimality. In an allocation model with heterogeneous patches, every positive allocation sees the same price; for an originally binding budget, a common linear utility term $s \sum_i x_i$ leaves the optimizer unchanged and shifts the price from p to $p + s$, as long as $p + s \geq 0$; and the price is zero when the budget is slack.
2. **A record window (Section 5).** In a toy universe with linear structure growth, a declared collapse threshold and erasure of records at turnaround, at least ten of twelve sampled patch populations retain records of both tag values through the horizon exactly when $-1/8000 < \lambda \leq \lambda_+$,

with $3 < \lambda_+ < 3.4$. The lower edge is set by the declared horizon; if records need only form eventually and then persist forever, the window of this gate becomes $[0, u_\infty)$ with $3 < u_\infty < 3.4$.

3. **What the readouts determine (Section 6).** In a self-consistent lattice model whose background offset is the analog of the constant term, the background does change the dynamics, but all eighteen sampled offsets converge. Fifteen pass a fixed sixty-update convergence test. Neither that test nor the existence of a limit recovers the offset. Adding a global density ledger makes the absolute background a gauge coordinate: it cannot be recovered from the invariant record, no selector that depends only on that record can follow a common shift, and no translation-invariant probability distribution over it exists.
4. **Typicality needs a prior (Section 7).** Five declared priors over the record window give different distributions of the coincidence ratio $R = |\lambda|/\rho_m$ at the first record; on the sampled grid none meets the declared criterion of putting at least 80% of its mass within one decade of the largest sampled ratio with a central 90% interval at most 1.5 decades wide. A smooth, positive prior can put almost all its mass on arbitrarily small ratios, so regularity near $\lambda = 0$ cannot by itself settle the coincidence.
5. **Audits of real data (Section 8).** On DESI DR2, Pantheon+ and compressed Planck data, a rewrite family $A(1+z)^m$ has a constant-compatible shape; the best-fit $w_0 w_a$ history crosses $w = -1$ at $z \approx 0.439$, where its density is positive, smooth and maximal; and a staging audit on supernova and BAO subsets, calibrated on mock catalogues, flags structured dependence between the subsets, led by the high-redshift supernova split. A separate audit of growth data finds that no single surveyed probe covers both regimes that a decisive test would need.

1.4 What "contingent modulus" means here

A parameter is a **modulus** of a model, relative to a declared readout, when that readout does not determine it: some two admissible configurations with different values give the same readout. It is **contingent** in the same relative sense: it must be supplied as input or chosen by some additional rule. The title's claim is meant in this sense and is established in two forms. In the original lattice model, two convergence readouts fail to determine the background offset (Theorem 6.2). In the ledger-closed extension, no function of the invariant record determines the absolute background, no selector depending only on that record transforms correctly under a common shift, and no translation-invariant probability distribution over the absolute value exists (Theorem 6.3; the last statement is a standard argument that is not mechanized).

Neither statement says that the physical cosmological constant is intrinsically undetermined, that no richer readout could fix the model parameter, or that no selection principle exists. They locate what a selection principle would have to supply in the models studied.

1.5 What this paper does not claim

The paper does not derive the value of Λ and does not refute a fundamental cosmological constant. It does not identify the closure price of Section 4 with the physical cosmological constant: that would require a map from the model's budget to a gravitational field equation with conserved stress-energy and the right dimensions and sign, which is not constructed here. The toy universe of Section 5 is a declared model, and its window is a property of its rules. The equation-of-state analysis of Section 8 shows that one expansion history has two regular descriptions; it does not decide whether physical dark energy is a fluid, and it says nothing about ghosts. The one real-data detection flags structured dependence between data subsets, consistent with a known tension; it is not a new-physics signal and does not identify the tension's source.

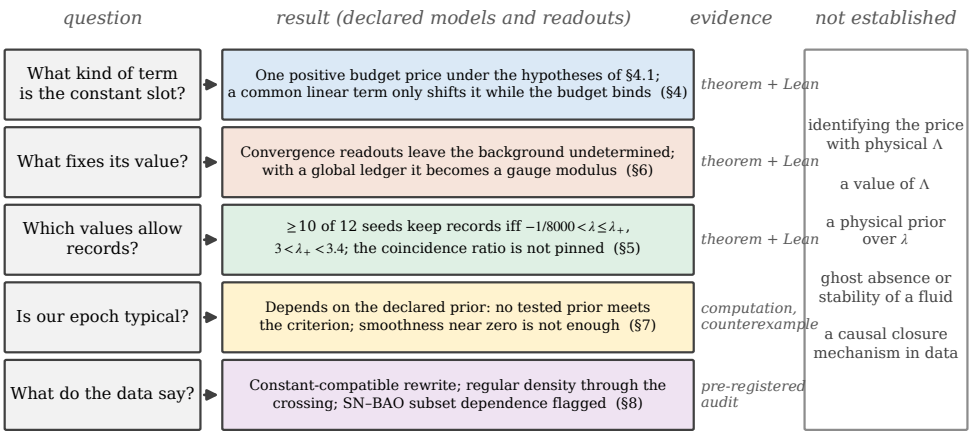


Figure 1. Overview. Each dark-energy question is posed to a finite model or a data audit. The middle column states what the paper establishes and the evidence type: a theorem (with Lean mechanization), a computation, or a pre-registered audit. All results are relative to the declared models and readouts. The right column lists what is not established.

2 The Six Birds vocabulary

This section introduces only the parts of the Six Birds closure calculus that the paper uses. Fuller accounts are in [32] and [31].

2.1 Theories as closure packages

The calculus describes theories by how they compress and complete a system. A **closure package** consists of a **lens** f, a deterministic coarse-graining that keeps some features of a substrate and discards the rest; a **completion** E, which fills in what the lens discarded whenever the description is used and is idempotent on its declared domain ($E^2 = E$); and **audits**, functionals that test whether the package stays consistent in use. The standard cosmological model can be read in these terms: the FLRW lens keeps a few homogeneous degrees of freedom, the homogeneous completion fills in the rest, and consistency tests between probes act as audits. We use this reading as a guide for building models, not as a theorem about cosmology.

The program organizes the moves a package can make into six primitive roles, P_1 to P_6. The Foundations series argues that these six are canonical under its composability and bounded-access assumptions [31]; that argument is not repeated or relied on here. The roles that appear in this paper are:

role	meaning	where it appears
P_1	operator rewrite	a correction term added to the macro law, such as the rewrite family $\Delta_H(z)$ (Section 8)
P_2	constraints and gating	admissibility rules: budgets, collapse thresholds, convergence tests (Sections 4–6)

role	meaning	where it appears
P_3	protocols and route structure	the order in which completions are applied, and probe routes in the data (Sections 6, 8)
P_4	staging and refinement	redshift bins, probe splits, finite iteration budgets (Sections 6, 8)
P_5	packaging	the quotient into macro-objects: patches, records, the homogeneous background
P_6	accounting	budgets and ledgers, whose multipliers are the shadow prices of Section 4

2.2 Readouts and determination

Many results in this paper answer the question *does this record determine that quantity?* Let $r : Z \to Y$ be a readout on a nonempty set Z of admissible configurations and $q : Z \to Q$ a quantity of interest. We say that r **determines** q if $q = \phi \circ r$ for some function $\phi : Y \to Q$, that is, if configurations with equal readouts always have equal values of q. If two admissible configurations have the same readout but different values of q, no decoder, however complicated, can recover q from r. This is the factorization test of the calculus. It is a statement about the readout, not about all possible observations: a richer readout may determine q.

A second test asks whether two completions commute, $E_1 E_2 = E_2 E_1$. A nonzero commutator means that the order in which two fill-in rules are applied matters. Section 6.5 applies both tests to a pair of coarse descriptions.

2.3 Budgets and shadow prices

The calculus records a conserved or budgeted quantity at one level as a **price** at the level above. If a description minimizes a cost J subject to a budget $e(x) \le B$, then under convexity and regularity conditions (Section 4.1) the multiplier of the budget is the rate at which the optimal cost $V(B)$ falls as the budget is relaxed, $p = -V'(B)$. Inverse temperature as the multiplier of a mean-energy constraint in entropy maximization, and prices in resource allocation, are familiar examples. What such a construction fixes is the *existence* and the *form* of the multiplier: one number per binding global constraint, zero when the constraint is slack. Its numerical value depends on the budget level and on the objective. Section 4 states the precise theorem used here.

In the Six Birds corpus this reading was also applied to holographic min-cut duality [27, 14]. The precise finite relation, which we use only as a comparison, is stated in Section 4.4.

2.4 Evidence types and provenance

We report every result with its evidence type:

- **Theorem [Lean]**: a mathematical statement about a declared model whose proof is checked by the Lean 4 kernel with mathlib [9, 29], using only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. Statements that rely on an argument not yet mechanized are labelled as such.
- **Exact computation**: a computation in rational or interval arithmetic outside Lean, with its inputs recorded.
- **Floating computation**: an ordinary numerical computation. Its reported precision is not a rigorous error bound.

- **Audit**: a pre-registered statistical test on public data, with its calibration and any amendments disclosed.

Results computed in this paper's repository are called **local**. Results taken from other Six Birds papers are called **imported**; they motivate the models and are never used as local proofs. One imported computation, the background sweep behind Section 6, was copied into the repository with a hash manifest, re-run, and re-derived by independent code; its rows are therefore local results, and Section 6 gives their mathematical analysis. Several cross-paper connections drawn elsewhere in the Six Birds corpus are not used, because their current source does not establish them as theorems: a blindness of the calculus to the Standard Model's generation count, an exclusion of proton decay and monopoles by record stability, and a null prediction for tabletop tests of gravitationally induced entanglement, which that source derives only for an assumed perfect-record channel.

3 Method

3.1 Three instruments

Theorems about declared models. Claims about the *form* of a term, about exact windows and about what a readout determines are theorems. Each is stated with its hypotheses. Most are mechanized in Lean 4; Appendix C lists the modules and the verification record. A theorem is only as relevant as its model, so each model is specified in full (Appendix A).

Computations on finite-dimensional models. Claims about behavior that is not yet proved, or about sampled quantities such as medians over seeds, come from running the models. Each model declares its state space, dynamics, swept parameter, grid, seeds and the predicate under test before the sweep. The record rule of Section 5 tests collapse, distinguishability and retention; it does not stipulate a target interval of λ. No observed cosmological value (Ω_Λ, H_0, 10^{-123}) appears in any model specification, predicate or threshold, and model quantities are reported in model units. The four models are the allocation model (Section 4), the record-window model (Section 5), the self-consistent lattice model (Section 6) and the measure probe built on the record-window model (Section 7).

Audits of public data. Claims about the observed universe are confined to Section 8. They are statistical audits of DESI DR2 BAO, Pantheon+ supernovae and compressed Planck distance priors. The staging audit of Section 8.3 is calibrated twice: on mock ΛCDM realizations through the same pipeline, to set its threshold, and on injected alternatives, to measure whether it can detect anything. The model comparison and the chart comparison are not calibrated in this way, the route residuals inherit the staging audit's calibration, and the growth diagnostics have their own injection controls.

3.2 Controls

The computed claims come with designated controls, alternatives on which the verdict is expected to change. A control that fails to change the verdict shows that the test has no demonstrated sensitivity to that alternative. The main controls are:

claim (section)	control and outcome
one price over heterogeneous allocations (4)	per-patch budgets of the same total give unequal prices

claim (section)	control and outcome
for an originally binding budget, a common linear term $s\,x_i$ leaves the optimizer unchanged provided the shifted price $p + s \geq 0$ (4)	without the global budget the same term moves the allocation; one patch moves by more than 1/2
the price is positive while the budget strictly binds (4)	budgets above the unconstrained total give price zero
records need structure (5)	with zero initial amplitudes no patch collapses at any λ, including $\lambda = 0$
records need both tag values (5)	constant tags fail the both-tag requirement even when patches collapse; in an independent-patch model the tag probability changes the probability of holding both tags
the limit readout carries information (6)	across the twelve sampled couplings, ten orbits converge and two approach two-cycles
the staging audit can detect a split (8)	probe-localized injections are detected in at least 80% of mocks at the largest tested amplitudes
the growth audit can detect a residual (8)	in the nonlinear model configurations a finite injected signal is detected; in the linear configurations the injection adds effectively no signal

The measure criterion of Section 7 has no positive control: no declared prior meets it, and Section 7.3 proves that on the base grid no reweighting of the grid values that keeps every value in the support can meet it.

3.3 Verification and reproduction

The original model and geometry-audit verdicts have build scripts, independently written validators that re-derive the verdicts from the raw output tables, and unit tests (Appendix C); additional computations have their own documented checks. Statements labelled as Lean theorems were checked in a full registered build of the Lean development, followed by registered declaration and type queries and transitive axiom checks confirming that the proofs use only the standard axioms (Appendix C); that each declaration states what the paper says it states is a matter of review. Statements argued on paper, and exact certificates computed outside Lean, are labelled as such where they occur. Agreement between two implementations, or between a proof and a computation, is a consistency check; it is not an independent human review.

The geometry audits do not depend on a Boltzmann solver: a compressed, scipy-based background pipeline is the only path they use. A pinned recipe recomputes the audit verdicts and repeats a reduced end-to-end fit within a declared tolerance (Appendix C).

3.4 Pre-registration and amendments

The real-data battery was pre-registered before its audit tables were generated. The registration fixed the models, splits, statistics and thresholds, and used data products and ΛCDM and $w_0 w_a$ fits that already existed from an earlier stage of the program; the registration therefore guards the audit computations, not first contact with the data. Two amendments were made during execution

and are logged in the registration record. First, the rewrite model was refitted through the same likelihood as the baselines after a first fit on a reduced data vector proved not comparable; that first fit is not reported as a result. Second, the staging audit's original class of injected alternatives, a rewrite applied coherently to all probes, was never detected, and was replaced by alternatives applied to one probe at a time (Section 8.3). A later mathematical review replaced the staging audit's interpolated percentile threshold with a conservative rank rule; this is a documented amendment, not a retroactive registration, and it does not change the verdict. Throughout, fitted amplitudes are in-sample parameters and are never reported as predictions.

4 The constant slot as the price of a global budget

The first question is what *kind* of term the constant slot is. In the closure calculus a global constraint at one level appears as a price at the level above (Section 2.3). This section makes that precise in three steps: a general theorem about when a budget carries a positive price, an adequacy residual that can serve as the budgeted quantity, and an allocation model in which the price shows the features usually attributed to a constant term. The comparison with holographic min-cut duality is stated at the end.

4.1 When a budget has a positive price

Let $\mathcal{A}$ be a convex subset of a real vector space, $J : \mathcal{A} \to \mathbb{R}$ a convex cost and $e : \mathcal{A} \to \mathbb{R}$ a convex budgeted quantity. The optimal cost at budget level B is

$$V(B) = \inf\{\, J(x) : x \in \mathcal{A},\ e(x) \le B \,\}.$$

Theorem 4.1 (Closure price)**.** *Suppose that $x^\star$ attains $V(B)$, that V is finite on a neighborhood of B, and that V is differentiable at B with $V'(B) < 0$. Then $p = -V'(B) > 0$, the budget binds, $e(x^\star) = B$, and*

$$J(x^\star) \ \le\ J(x) + p\,(e(x) - B) \qquad \text{for every } x \in \mathcal{A}.$$

In particular $x^\star$ minimizes the Lagrangian $J + p\,(e - B)$ over all of $\mathcal{A}$, the primal and dual optimal values agree, and any other certificate with these properties has the same price.

The theorem constructs the price from the value function itself; no Lagrange multiplier, constraint qualification or convexity of V is assumed. It is mechanized in Lean, together with several variants (Appendix C): objectives taking the value $+\infty$ outside an effective domain, matrix-valued budgets $E(x) \preceq \mathcal{B}$ with a positive semidefinite matrix price, prices obtained from a Slater point without any differentiability of V, and a full stationarity condition split into objective, budget and domain terms under a relative-interior qualification.

Two hypotheses deserve comment. A binding budget alone does not give a positive price: minimizing x^2 subject to $x \le 0$ makes the constraint active at $x = 0$, Slater's condition holds, and the multiplier is zero. Strict sensitivity, $V'(B) < 0$, is what forces positivity, and the theorem shows that it also forces the budget to bind. Second, the theorem fixes the *existence and form* of the price, not its value, which depends on B and on J.

4.2 An adequacy residual as the budgeted quantity

In the calculus the natural budgeted quantity is the part of one family of descriptions that another family cannot explain. For whitened finite matrices L (native probes) and D (dissolving probes), let

P_L be the orthogonal projector onto the row space of L, built with the Moore–Penrose inverse. The **adequacy residual** is

$$\Xi \;=\; D\,(I - P_L)\,D^\top \;\succeq\; 0.$$

Proposition 4.2 (Adequacy residual)**.** *For every native rank, including singular $LL^\top$: (i) $\Xi = 0$ if and only if $D = AL$ for some matrix A, if and only if $\ker L \subseteq \ker D$; (ii) for every linear explanation A, $(D - AL)(D - AL)^\top = \Xi + (A - A_\star)LL^\top(A - A_\star)^\top$ with $A_\star = DL^+$, so Ξ is the Loewner-smallest residual; (iii) for fixed L and affine $D(x)$, the map $x \mapsto \Xi$ is Loewner-convex, with derivative $H(I - P_L)D^\top + D(I - P_L)H^\top$ in direction H.*

All three parts are mechanized. Part (iii) makes $\mathrm{tr}(W\Xi)$, for a positive semidefinite audit weight W, a convex budgeted quantity to which Theorem 4.1 applies; the mechanized examples include a complete Schur-budget instance with its optimizer, value and derivative. If W is only semidefinite, a zero trace need not mean $\Xi = 0$. When the native probes vary as well, convexity can fail, and a mechanized example shows this.

4.3 An allocation model

The theorem says when a price exists. To see what the price *does*, consider $N = 32$ patches with allocations $x_i \geq 0$, each with a concave local utility

$$u_i(x_i) = w_i \log(1 + x_i) - \tfrac{1}{2}c_i x_i^2 + b\,x_i,$$

and one global budget $\sum_i x_i \leq B$. The weights w_i and curvatures c_i are fixed, deliberately heterogeneous arrays (Appendix A.1), and b is a common linear offset, zero unless stated otherwise. For every finite family with $w_i, c_i > 0$, every offset and every $B \geq 0$, the problem has a unique optimizer, constructed explicitly, with a budget price $p(B)$ that is unique for $B > 0$ and nonincreasing in B. Writing the optimal *cost* as $V(B) = -\max \sum_i u_i(x_i)$, V is continuously differentiable for $B > 0$ with $V'(B) = -p(B)$, including where the set of active patches changes. These statements are mechanized, and the original 32-patch inputs are reified exactly in Lean. Three properties follow.

One price over unequal allocations. At the optimum every patch with $x_i > 0$ satisfies $u_i'(x_i) = p$, and every patch with $x_i = 0$ satisfies $u_i'(0) \leq p$. In the original inputs at $B \approx 22.24$ all allocations are positive and differ from patch to patch (0.735, 0.800, 0.828, 0.878, …), while the price is one number, $0.46112 < p < 0.46113$. The heterogeneity is essential: with identical patches the uniformity would be trivial. *Control:* replacing the global budget by per-patch budgets with the same total gives unequal per-patch prices (0.494, 0.548, 0.568, …). A single price comes from the globality of the constraint. It does not mean that the price ignores local detail: changing the local weights at the same budget changes the price.

A common offset shifts the price, not the allocation. Adding the same linear term $b\,x_i$ to every patch adds $b\sum_i x_i = bB$ to the objective on the binding face of the budget. For an active budget the optimizer is therefore unchanged and the price becomes $p + b$, provided $p + b \geq 0$ so that the budget still binds; for the original inputs this holds for every $b \in [-\frac{1}{4}, \frac{1}{4}]$. A literal constant added to the objective would change neither the optimizer nor the price; the linear offset is the analog of a constant energy density times a volume share. In floating point at $b = \pm 0.2$ the allocation moved by at most 5.7×10^{-14} and the price shifted by b to within 3.3×10^{-14}. *Control:* without the global budget the same offset moves the allocation, by 0.56 to 0.67 in floating point; the exact proof certifies a displacement larger than $1/2$ for one patch. The offset is screened only because the budget binds.

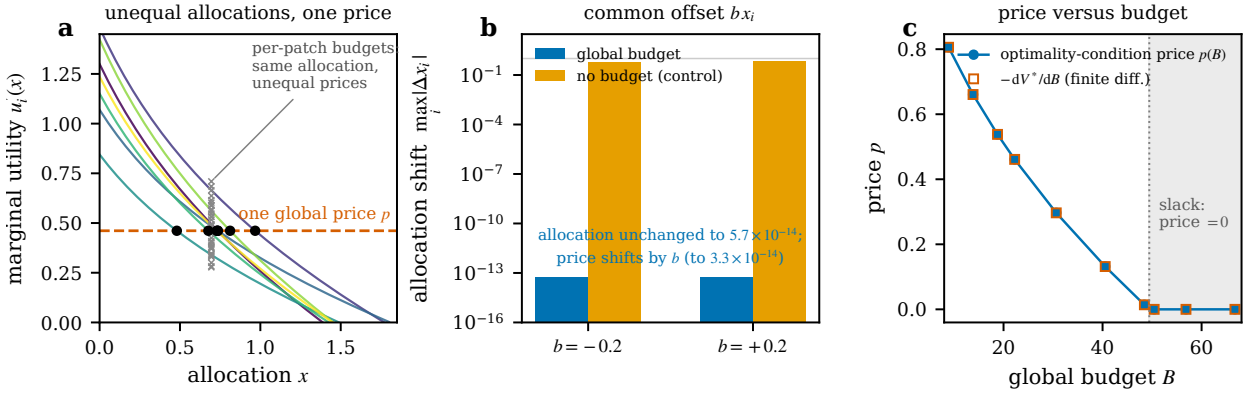


Figure 2. The allocation model. *Left:* marginal utilities of several heterogeneous patches meet one global price at their optimal allocations (dots); in the per-patch-budget control every patch receives the same allocation but the prices differ (grey crosses). *Center:* a common linear offset $b\,x_i$ with $b = \pm 0.2$ leaves the allocation unchanged to floating-point precision when the global budget binds, while the price shifts by b; without the budget the same offset moves the allocation. *Right:* the price from the optimality conditions agrees with minus the finite-difference derivative of the optimal cost and is zero once the budget is slack. Floating-point outputs; the corresponding exact statements are proved in Lean.

The price vanishes when the budget is slack. Across the ten budgets of the sweep, seven bind with positive price and three are strictly slack with price exactly zero. While the budget binds, the price falls as the budget grows (0.806, 0.661, 0.538, 0.461, 0.297, 0.132, 0.014), and it agrees with minus a central finite difference of the optimal cost to 3.6×10^{-11}, the largest discrepancy in the sweep.

Together these are the familiar features of a constant term: one value shared by all patches, insensitivity of the state to a uniform offset, and absence when the constraint is not active. In unimodular gravity the cosmological constant appears as an integration constant decoupled from vacuum energy [16], and in vacuum-energy sequestering a global constraint with a Lagrange multiplier removes the Standard-Model vacuum energy from the gravitational dynamics [19]. The allocation model shows the same pattern in a finite, checkable setting. It does not show that physical vacuum energy decouples.

4.4 Comparison with min-cut duality

The Ryu–Takayanagi prescription relates boundary entanglement entropy to a bulk minimal area [27], and in discrete flow formulations max-flow/min-cut is a linear-programming duality [14]. Here we compare the closure price with a declared finite min-cut model. The precise finite relation is the following. For capacities c, the minimal cut cost $A(c)$ is the optimal value of the dual problem. A minimizing cut supplies a supergradient of A whose components are its cut-incidence coefficients; where the minimizing cut is unique and stable, these are the marginal prices of the individual edge capacities, and they are not the area. For a fixed capacity shape c_0 and a common scale $s \geq 0$, however, $A(s\,c_0) = s\,A(c_0)$, so the area *is* the marginal price of the common scale at $s = 1$. A slack regime with zero price exists only if a throughput ceiling is added. The finite-minimum, weak-duality, tight-certificate and common-scale laws are mechanized; on the 14-edge graph used in the Six Birds corpus, the minimizing cuts, their stability margins and the individual capacity prices are checked in exact rational arithmetic (254 boundary masks, 8{,}128 cut–flow inequalities). Area and closure price are therefore both outputs of an optimization duality, but they play different roles: one is an

optimal value under a declared scaling, the other a multiplier. We use the comparison as an analogy only. A proposed rule under which entanglement entropies compose like cut areas fails on an explicit, normalized two-qubit contraction, whose entropy drops from $H(3/4)$ to $H(9/10) < H(3/4)$.

4.5 Summary

Under the attainment, finiteness, convexity and sensitivity hypotheses of Theorem 4.1, a global budget carries one positive scalar price. For the separable allocation problem with the budget $\sum_i x_i \le B$, that price is shared by every patch with a positive allocation; for an originally binding budget, a common linear utility term $b\,x_i$ leaves the optimizer unchanged and changes the price to $p+b$, provided $p+b \ge 0$; and the price is zero when the budget is slack. These are statements about optimization problems. Whether the physical cosmological constant is such a price is a separate question, which would require a map from the budget to a gravitational field equation; it is not addressed here. The next two sections ask what the models say about the range of admissible values and about the value itself.

5 Which values allow records: an exact window

A constant term enters the expansion history, and through it the growth of structure. This section asks a question that does not require knowing the value: in a simple universe that forms structure, which values of the constant allow lasting, distinguishable records? The answer is an exact window for a declared toy model. The selection predicate never mentions λ.

5.1 The model

Units are chosen so that the matter density is $\rho_m = a^{-3}$ and $H^2 = a^{-3} + \lambda$; the swept parameter λ is a signed constant. Each of $N = 64$ patches carries a linear density contrast $\delta_i(a) = m_i\, D_\lambda(a)$, where D_λ solves the standard linear growth equation $\ddot{D} + 2H\dot{D} = \frac{3}{2}\rho_m D$ from $a_0 = 10^{-3}$ with $D(a_0) = 1$ and $dD/da(a_0) = 1/a_0$. The amplitudes m_i are drawn once per seed from a half-normal distribution of scale 10^{-3}, and each patch carries an independent Boolean tag. The rules are:

- **formation:** a patch collapses when its contrast reaches $\delta_c = 1.686$ [15], and its tag is frozen as a record;
- **erasure:** if the expansion turns around at or before the horizon $A = 20$, which happens exactly when $\lambda \le -A^{-3} = -1/8000$, all records are erased; this is an imposed rule standing in for recollapse;
- **seed predicate:** a seed passes if, by $a = A$, collapsed patches carrying *both* tag values exist and have not been erased;
- **aggregate gate:** a value of λ is admitted if at least 10 of the 12 seeds pass (the declared 80% gate).

Two facts about the model drive everything below. On the expanding branch, which exists through $a = A$ exactly when $\lambda > -1/8000$, the growth factor $D_\lambda(a)$ is strictly increasing in a and, for every $a > a_0$, strictly decreasing in λ: more dark energy means less growth. And a seed j passes exactly when

$$\lambda > -1/8000 \quad \text{and} \quad G(\lambda) = D_\lambda(A) \ge T_j = \max\big(\delta_c/m_{j,0},\, \delta_c/m_{j,1}\big),$$

where $m_{j,0}$ and $m_{j,1}$ are the largest amplitudes carrying each tag (with $T_j = \infty$ if a tag is absent). The aggregate gate is then $\lambda > -1/8000$ and $G(\lambda) \geq T_\star$, where $T_\star$ is the tenth smallest of the twelve seed thresholds.

5.2 The window

Theorem 5.1 (Record window)**.** *For the realized twelve-seed, 64-patch population, the set of admitted λ is exactly*

$$(-1/8000,\ \lambda_+], \qquad 3 < \lambda_+ < \tfrac{17}{5},$$

where λ_+ is the unique solution of $G(\lambda_+) = T_\star$ and

$$T_\star = \frac{971912828383572000768}{1012057850670495125} \approx 960.333.$$

The proof is mechanized end to end: existence and uniqueness of the growth solution for every admissible λ, positivity and strict monotonicity in a and in λ, continuity and differentiability in λ, the limit $G(\lambda) \to \frac{3}{2} - \frac{1}{2}(a_0/A)^2$ as $\lambda \to \infty$, the exact reification of all 768 realized amplitudes and tags, the twelve thresholds and their rank, and a quadrature certificate for the bracket on λ_+. Because $G(0) = A/a_0 = 20000$ and G decreases, every negative admissible λ passes. The lower edge is therefore set entirely by the erasure rule and the horizon, $-A^{-3}$; the upper edge is set by growth.

The original sampled sweep agrees. On a signed grid spanning eight decades in $|\lambda|$, the 80% gate passed between the inside and outside witnesses -1.18×10^{-4} and -2.7×10^{-4} at the lower edge and 2.33 and 5.31 at the upper edge (Figure 3a). The two outside witnesses fail for different reasons. At $\lambda = -2.7 \times 10^{-4}$ exactly 58 patches of seed 1100 collapse before turnaround, with both tags present, and the erasure rule then leaves none; this count is certified from the exact solution. At $\lambda = 5.31$ only one patch of that seed collapses, so no pair of distinguishable records forms. A homogeneous control with all amplitudes zero forms no records anywhere, including at $\lambda = 0$.

The window is a property of these rules. The lower edge moves with the horizon, and its location carries no meaning beyond the declared persistence requirement.

5.3 Variants

Eternal records. If records need only form at some finite time and then persist forever, every negative λ eventually turns around and is excluded, and the window becomes $[0, u_\infty)$ with $\lambda_+ < u_\infty$ and $3 < u_\infty < 3.4$. The upper end is open: at u_∞ exactly nine seeds eventually obtain both tags, one short of the gate. Requiring formation by $a = 20$ and permanent retention instead gives the closed window $[0, \lambda_+]$. The finite-horizon endpoints converge to u_∞ as the horizon grows.

Population size. For the separately realized populations with $N = 32$ and $N = 128$ patches, the windows are strictly nested: $\lambda_+^{(32)} \in (1.3, 1.7)$, $\lambda_+^{(64)} \in (3, 3.4)$ and $\lambda_+^{(128)} \in (5, 6.25)$, with the same lower edge (Figure 3b). More patches produce more high-amplitude outliers, which still collapse at larger λ. This is a theorem about these three populations, not a monotonicity law for arbitrary reseeding. For all three, $\lambda \geq 32$ is a sufficient cutoff for permanent starvation: growth stays below 485 at every finite epoch, while every patch needs more than 485 to collapse.

Threshold and amplitude scale. Under an exact common rescaling of the amplitudes by $s \in [\frac{1}{3}, 3]$ and any threshold $\delta_c \in [1.2, 2]$, the window exists throughout, and its upper edge is continuously differentiable, increasing in s and decreasing in δ_c.

Tags. The seed predicate requires both tag values, so the tag distribution matters: constant tags fail it even when individual patches collapse. Suppose the patches are independent and identically distributed, each collapsing with probability p and carrying tag 1 with probability q, with tag and

collapse independent within a patch. Then a population of N patches holds collapsed patches of both tags with probability

$$P_N = 1 - (1 - pq)^N - (1 - p(1 - q))^N + (1 - p)^N,$$

which is nondecreasing in N and depends on q; for $N = 2$ and $p = 1$ it is $1/2$ at $q = 1/2$ and $3/8$ at $q = 1/4$. The sampled control with $q \in \{0.45, 0.50, 0.55\}$ left the gate unchanged on the coarse grid, which reflects how close these values are, not an irrelevance of the tags.

5.4 The coincidence ratio is not pinned

The strong form of the coincidence problem asks whether record-bearing universes must find the constant term comparable to matter when records form. Define the ratio at the first record,

$$R(\lambda) = \frac{|\lambda|}{\rho_m(a_{\mathrm{rec}})},$$

where a_{rec} is the first epoch at which any patch of the seed collapses.

Proposition 5.2 (No pinch)**.** *For every seed of the realized population: throughout the window,* $0 \le R < 425/8$*; for admitted* $\lambda < 0$*,* $0 < R < 1/8000$*; at the upper edge,* $3/8 < R(\lambda_+) < 425/8$*; and on* $0 \le \lambda \le \lambda_+$ *the first record forms at* $1/2 < a_{\mathrm{rec}} \le 5/2$*. Moreover, for every* $\varepsilon > 0$ *there is an admitted* $\lambda > 0$ *at which all twelve ratios are below* ε*.*

So the window bounds the ratio above, with order-unity values at the upper edge, but it does not force the ratio toward unity: arbitrarily small positive ratios occur at admitted values of λ approaching zero. The sampled sweep shows the same picture. Across positive admitted grid values the median over seeds ranges from 3.5×10^{-6} to 2.1, about 5.8 decades (Figure 3c); individual seed ratios at nonzero admitted grid values range from 1.3×10^{-6} to 17.4. The shape resembles Weinberg's anthropic bound [35]: an upper limit on λ from the requirement that structure forms, without a prediction that the observed value sits near that limit. Whether our epoch is typical depends on how values inside the window are weighted, which is the subject of Section 7.

6 What determines the value

Section 4 concerned the form of the constant slot and Section 5 the range of values compatible with records. This section asks what a model's own dynamics and records determine about the value. We use a small self-consistent lattice model in which a background offset plays the role of the constant term, classify its orbits exactly, and then ask which readouts recover the offset. A closed-ledger extension of the model shows how the price construction of Section 4 turns the absolute value into a gauge coordinate.

6.1 The lattice model

The model comes from the Six Birds corpus; its source files were copied into this repository with a hash manifest, re-run, and re-classified by independently written code with exact agreement on every row (Appendix C). A complex field ψ on an eight-site ring sources the potential it moves in:

$$T_{00,i}[\psi] = |(\nabla\psi)_i|^2 + V_{\mathrm{bg},i}\,|\psi_i|^2, \qquad V_i[\psi] = V_{\mathrm{bg},i} + \kappa\, T_{00,i}[\psi].$$

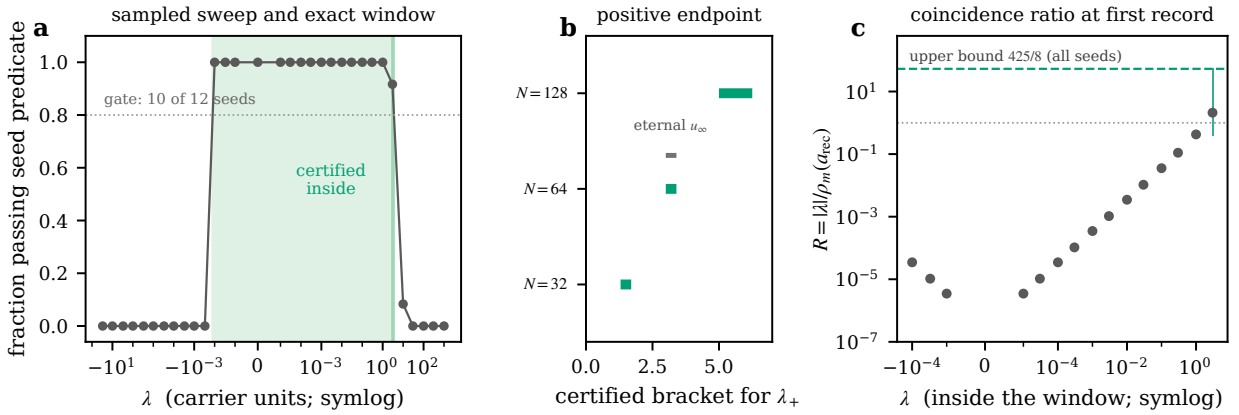


Figure 3. The record window. *(a)* Fraction of the twelve seeds passing the both-tag seed predicate in the original sampled sweep (grey). The admitted window is $(-1/8000, \lambda_+]$: the light green region $(-1/8000, 3]$ is certified to lie inside it, and the darker band is the certified bracket $3 < \lambda_+ < 3.4$ for its upper edge. *(b)* Certified brackets for the upper edge of the three separately realized populations, and for the eternal-record edge u_∞ of the 64-patch population. *(c)* Seed-median coincidence ratio at the first record in the sampled sweep (grey) with the certified bounds: $R < 425/8$ throughout the window for every seed (dashed) and $3/8 < R(\lambda_+) < 425/8$ at the upper edge (green box; its horizontal extent is the bracket for λ_+). The ratio can also be made arbitrarily small at admitted $\lambda > 0$.

One update takes the normalized lowest eigenvector ϕ of $H = K + \mathrm{diag}(V[\psi])$, where K is the ring's kinetic operator, replaces it by $\phi\, e^{-i \arg z}$ with $z = \langle \phi, \psi \rangle$ (unless $|z|$ is below a small tolerance), and sets $\psi \leftarrow (\psi + \phi)/\|\psi + \phi\|$ (mixing 1/2). This is the source's phase convention, which we keep unchanged. A run is declared converged if, after 60 updates, four diagnostics are all at most 10^{-8}: the size of the last update, the fixed-point gap between ψ and the lowest eigenvector of the final Hamiltonian after the source's phase operation is applied to it twice, the eigen-equation residual of ψ, and a sourcing residual. As implemented, the sourcing residual compares the potential with itself and is identically zero, so in effect three diagnostics are tested. The background is a fixed site profile plus a scalar offset c. The original sweep used eighteen offsets in $[-8, 10]$ at coupling $\kappa = 0.3$, and twelve couplings from 0 to 30 at zero offset.

The statements below about the original sampled orbits and diagnostics refer to exact real and complex trajectories on the rational values of the saved binary inputs: the initial field, the background arrays (including each shifted array as stored), the couplings and the thresholds. Decimal labels such as $\kappa = 0.3$ or offset -5 name those inputs. The statements do not describe trajectories for idealized decimal inputs or the rounding of each floating-point operation.

The offset is not a symmetry of the dynamics. Shifting the background by c changes the potential at site i by

$$\Delta V_i = c\,(1 + \kappa |\psi_i|^2),$$

which is not constant across sites unless the density is uniform. Different offsets therefore give genuinely different self-consistent states, and a readout that kept the whole state could in principle tell offsets apart.

6.2 Finite budgets and limits

In the original sweep, fifteen of the eighteen offsets passed the 60-update test and three, $\{-5, -3, +10\}$, failed; among the couplings, $\kappa \le 10$ passed and $\kappa \in \{15, 20, 30\}$ failed. A pass/fail table of this kind

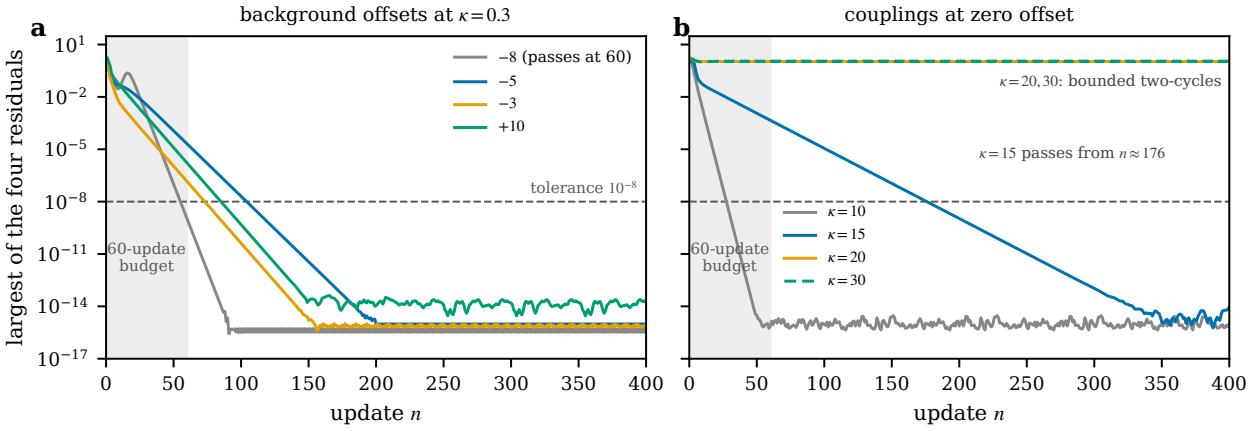


Figure 4. Finite budget versus limit in the lattice model. Largest of the four convergence residuals after n updates, computed in floating point with the unchanged original code. *Left:* offsets -5, -3 and $+10$ fail the test at the 60-update budget (shaded) but pass it later; offset -8 is shown for comparison. *Right:* coupling $\kappa = 15$ first passes at update 176; $\kappa = 20$ and 30 approach bounded two-cycles. The exact statements, including convergence, are in Theorem 6.1.

is easily read as a map of which parameter values *admit* a self-consistent state. The exact analysis shows that it is not.

Theorem 6.1 (Orbit classification)**.** *For the original eight-site model, initial field and mixing* $1/2$*:*

1. *every real background profile and every real coupling admit a positive, normalized, self-consistent ground state;*
2. *at* $\kappa = 0.3$*, the orbits of all eighteen sampled offsets converge to a self-consistent ground state; fifteen pass the four-residual test at update 60, and the offsets* -5*,* -3 *and* $+10$ *fail it there;*
3. *at zero offset, the orbits for* $\kappa \in \{0, 0.1, 0.3, 0.5, 1, 2, 3, 5, 10\}$ *converge and pass the test at update 60; the orbit for* $\kappa = 15$ *fails the test at every update from 1 to 60, yet converges, entering a certified basin of attraction at update 178; the orbits for* $\kappa = 20$ *and* $\kappa = 30$ *approach bounded two-cycles, so they have no limit.*

Each part is mechanized. Part 1 uses Brouwer's fixed-point theorem for the self-consistency map and does not assume that any iteration converges; a smooth positive ground branch is also constructed along the original background for every coupling in $[0, 30]$. Parts 2 and 3 follow the exact orbits from the original seed: through finite stretches by exact arithmetic, and from there by certified contraction into the limit, or into the attracting two-cycle. All sampled orbits stay normalized and within the update's legality guards at every step. The classifications concern the sampled parameter values; they do not cover every intermediate offset or coupling.

Figure 4 shows the same behavior in ordinary floating point. The three "failing" offsets -3, $+10$ and -5 first pass the diagnostics at updates 74, 86 and 105, the coupling $\kappa = 15$ at update 176, and the couplings 20 and 30 settle into oscillations whose residuals stay of order one. The 60-update test therefore measures how fast an orbit converges, not whether a self-consistent state exists. Nor does it separate the offset from the coupling as kinds of parameter: both have sampled values that fail at 60 updates and converge later. The coupling differs only in having sampled values, 20 and 30, whose orbits have no limit even though self-consistent states exist there.

6.3 Readouts that do not determine the offset

Two Boolean readouts are natural for this model: whether the four-residual test passes at update 60, and whether the orbit has a limit at all.

Corollary 6.2 (Readout nonidentifiability)**.** *Let $r_{60}(c)$ be the outcome of the four-residual test at update 60 and $r_\infty(c)$ the truth value of "the orbit has a limit", for the eighteen sampled offsets c at $\kappa = 0.3$. No function $\phi : \{\text{true}, \text{false}\} \to \mathbb{Q}$ satisfies $\phi(r_{60}(c)) = c$ for all eighteen offsets, and none satisfies $\phi(r_\infty(c)) = c$. Concretely, $r_{60}(-8) = r_{60}(-2) = \text{true}$, and r_∞ is true for every offset.*

Given Theorem 6.1, the corollary is elementary, and it is mechanized with the readouts defined from the actual orbit propositions rather than from a table. Its content is the precise sense in which the offset is a *contingent modulus* of this model: the two convergence readouts leave it undetermined, so it must be supplied as input or chosen by some further rule. Three limits of the statement matter. The finite readout does distinguish passing from failing offsets. A richer readout, such as the full converged state, could recover the offset, since the dynamics depend on it. And nothing here excludes an additional selection rule.

6.4 A closed ledger makes the absolute value a gauge coordinate

The offset in the lattice model is visible to the full state because it couples to the density through κ. Section 4 suggests what would hide it: a global budget whose price absorbs a common shift. The sitewise shift $c\,(1 + \kappa\rho_i)$, with $\rho_i = |\psi_i|^2$, is exactly the derivative with respect to ρ_i of $c\,Q_\kappa$, where

$$Q_\kappa(\psi) = \sum_i \Big(\rho_i + \tfrac{\kappa}{2}\rho_i^2\Big),$$

and among quadratic ledgers $a\rho + b\rho^2$ this one is forced. The added force $c\,(1 + \kappa\rho_i)\psi_i$ is therefore normal to the level sets of Q_κ. Fixing Q_κ adds a constraint to the model, with a multiplier μ. The compensated potential

$$V_i^{\text{closed}} = V_{\text{bg},i} + \kappa T_{00,i} - \mu\,(1 + \kappa\rho_i) = (V_{\text{bg},i} - \mu)(1 + \kappa\rho_i) + \kappa\,|(\nabla\psi)_i|^2$$

depends on the background only through $V_{\text{bg}} - \mu$, so it is unchanged under $(V_{\text{bg}}, \mu) \mapsto (V_{\text{bg}} + c, \mu + c)$ for every field. A presentation of the closed model is a triple (state, background, price); its invariant record keeps the full state and every relative background coordinate $V_{\text{bg},i} - \mu$.

Theorem 6.3 (Gauge modulus)**.** *In the ledger-closed model:*

1. *two presentations have the same invariant record if and only if they differ by a common shift of background and price;*
2. *the absolute price, equivalently the absolute background, is not a function of the invariant record; in the allocation form of the closed model this holds already on an explicit pair of two-site optimality certificates, with $\kappa = 2$, densities $(1/2, 1)$, ledger budget $11/4$ and positive prices $3/2$ and $5/2$;*
3. *no selector depending only on the invariant record transforms correctly under the joint shift $(V_{\text{bg}}, \mu) \mapsto (V_{\text{bg}} + c, \mu + c)$, since the shift leaves the record unchanged while an absolute value would have to move by c;*
4. *any linear projection that annihilates the ledger normal makes the projected force at a given state invariant under the common shift; and in the allocation form of the closed model, two projected-gradient histories with the same initial state, budget, step size and coefficients, whose backgrounds differ by a common shift, coincide at every step.*

In addition, no probability measure on the line of absolute values is invariant under all real translations.

Parts 1 to 4 are mechanized as stated. Invariance of the continuous-time constrained flow under the shift follows from part 4 together with the existence and uniqueness argument below, and is not mechanized. The final statement is the standard argument that translation-invariance would give every unit interval the same mass, which is impossible for a probability measure; it is not mechanized. The extension changes the model: it adds the ledger and replaces the damped eigenvector update by a constrained flow, with the normalization and $I = \sum_i \rho_i^2$ held fixed, that keeps the original local force. That the projection onto this constraint set is well defined and the flow exists globally is a paper-level argument, not mechanized: for $\kappa > 0$, on the level set $\sum_i \rho_i = 1$, $I = 17/108$, which is nonempty and compact, the two constraint normals are independent (because $I \neq 1/m$ for $m = 1, \ldots, 8$), so the projected field is smooth and tangent, and solutions exist and are unique for all times.

The theorem gives a sharper sense to the title in this model. The absolute value is not merely unread by a coarse test; it is a gauge coordinate of the closed description. A convention such as $\mu = 0$, or any rule that picks a value for outside reasons, remains available. What is excluded is recovering the value from the invariant layer itself, and assigning it a translation-invariant probability. Any prior over the absolute value therefore has to carry symmetry-breaking content, a point that returns in Section 7. Nothing here shows that the physical cosmological constant is a gauge coordinate of any physical theory.

6.5 Two descriptions and a mediator

The headline form of the cosmological-constant problem compares a vacuum-energy estimate made in quantum field theory with a constant measured through gravity. In the Six Birds corpus this comparison is framed as one between two coarse descriptions of a common finer system. A finite version of that framing has a precise theorem, which we record because it states what any common description would have to retain.

Let $H = \{0,1\}^4$ with coordinates (d_0, d_1, d_2, d_3), and consider the two readouts $q_Q(d) = (d_0, d_1, d_2)$ and $q_G(d) = (d_0, d_2, d_3)$. Neither determines the other.

Theorem 6.4 (Mediator)**.** *A deterministic map r on H, with any codomain, determines both q_Q and q_G if and only if it determines the full state d, which holds if and only if r is injective. In particular such a mediator has exactly sixteen distinct records.*

The theorem is mechanized, and the recovery map is explicit: three coordinates from the first decoder and the fourth from the second. It concerns information, not storage: it does not require four separate fields or four dimensions. The companion question, whether the two fill-in rules commute, has an affirmative answer for the original model: the completions given by uniform conditional means over the two readouts commute on every observable. A separately declared weighted example does not commute: on a three-cell block with cell weights 5, 1 and 3, applying the two conditional means in opposite orders to the observable $(0, 1, 0)$ gives values that differ by $1/16$ on the first cell.

For the cosmological-constant problem the lesson is limited and structural. Comparing a number computed in one description with a parameter of another requires a map between them, and in this finite model any map that supports both readouts must retain everything either one sees. The theorem does not identify the two readouts with quantum field theory and gravity, does not compute a vacuum-energy discrepancy, and does not resolve the cosmological-constant problem.

7 Is the present epoch typical? Measures over the window

Section 5 showed that the record window bounds the coincidence ratio $R = |\lambda|/\rho_m(a_{\rm rec})$ from above but does not pin it near unity. Typicality is a further question: it needs a way to weight values of λ inside the window. Neither model of Sections 5 and 6 supplies such a weighting, and Section 6.4 showed that in the ledger-closed model no translation-invariant one exists. So a measure must be declared. This section declares five, computes what each implies for R, and then proves two general limits on what any argument of this kind can conclude.

7.1 The probe

On the record-window model of Section 5, a denser grid places 36 logarithmically spaced positive values of λ inside the window, together with a sliver of negative values and $\lambda = 0$. For each value and each of the twelve seeds we record whether the seed passes, its number of distinguishable records $N_{\rm rec}$, its collapsed fraction $f_{\rm coll}$ [25], its first-record epoch $a_{\rm rec}$ and the epoch a_λ at which $\rho_m(a_\lambda) = |\lambda|$. Five measure classes were declared before any verdict was computed:

class	weight	rationale
M1	flat in λ	the linear prior behind the classic typicality argument
M2	flat in $\log\lambda$	the scale-invariant alternative
M3	flat in λ, times $N_{\rm rec}$	each record counted as one observer-analog
M4	flat in λ, times $f_{\rm coll}$	a collapsed-fraction weight, as in [22]
M5	flat in $\log\lambda$, times $N_{\rm rec}$	whether record weighting can concentrate a logarithmic prior

Each grid value receives the mass its class assigns to its cell. That mass is shared equally among the twelve seeds, and each seed that passes contributes its own value of R with its share multiplied by its own record weight. Quantiles are taken from the resulting weighted empirical distribution. Logarithmic classes use the declared positive cutoff 10^{-5}, and $\lambda = 0$ is excluded from distributions of $\log R$. The pre-registered criterion says that a class *concentrates* R if the probability that $\log_{10} R$ lies within one decade of its largest supported value is at least 0.8, and the central 90% interval of $\log_{10} R$ is at most 1.5 decades wide. The expectation recorded in advance was that linear classes with record weights would concentrate and logarithmic ones would not.

7.2 Results

The table and Figure 5 are floating-point computations on the sampled grid, using numerically integrated growth histories.

class	P(top decade)	90% width of $\log_{10} R$ (decades)	median $a_\lambda/a_{\rm rec}$	meets criterion
M1 flat λ	0.331	1.82	1.36	no
M2 flat $\log\lambda$	0.041	4.96	8.71	no
M3 λ, $\times N_{\rm rec}$	0.194	2.22	1.71	no
M4 λ, $\times f_{\rm coll}$	0.194	2.22	1.71	no
M5 $\log\lambda$, $\times N_{\rm rec}$	0.009	4.42	13.80	no

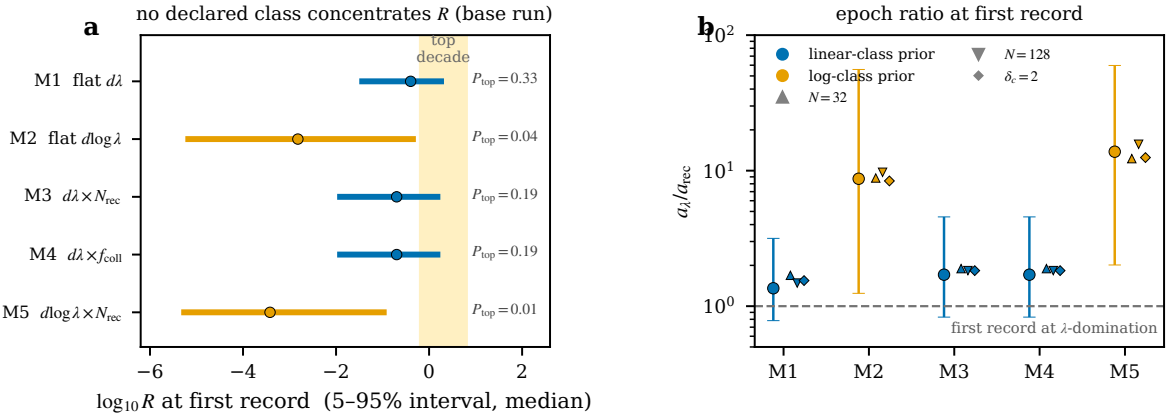


Figure 5. Declared measures over the record window. *(a)* Central 90% interval and median of $\log_{10} R$ for each class in the base run, with the probability of the top decade (shaded). *(b)* Median and 90% interval of the epoch ratio $a_\lambda/a_{\rm rec}$ in the base run, with the medians of the three robustness variants. Blue: classes flat in λ; orange: classes flat in $\log\lambda$.

No class meets the criterion, and none does in any of the three declared robustness variants ($N = 32$, $N = 128$, $\delta_c = 2$); the largest top-decade probability among them is 0.68, for M1 at $N = 128$. Classes M3 and M4 coincide exactly, because on every passing seed $f_{\rm coll} = N_{\rm rec}/N$ with N fixed, so the two weights are proportional.

Two comparisons survive the negative verdict. Linear classes place the typical first record close to the epoch of λ-domination, with median $a_\lambda/a_{\rm rec}$ between 1.36 and 1.71, while logarithmic classes place it an order of magnitude earlier. And record weighting does not help concentration: it *broadens* the linear distribution (top-decade probability $0.33 \to 0.19$, width $1.82 \to 2.22$ decades), because record counts fall as growth is suppressed at large λ, and it does not rescue the logarithmic prior. The top decade is measured from the largest supported seed value of R, so it is sensitive to rare seeds; it was not adjusted after the fact. No measure built from the model's own structure was constructed, so the comparison is among declared classes only.

7.3 What a prior must supply

The computation above uses a frozen grid. Two theorems show that the conclusion does not hinge on the particular classes.

Proposition 7.1 (Fixed-support bound)**.** *On the frozen base grid, with every admitted grid value kept in the support and seeds weighted equally, every prior that reweights only the grid values gives a top-decade probability of at most* $8/11 < 4/5$*; with record weights the bound is* $57/80$*. Both bounds are sharp.*

The bound comes from the fact that, at each grid value, at most eight of the eleven or twelve passing seeds fall in the top decade, and a reweighting of grid values only mixes these conditional fractions. The algebra is mechanized; the counts are checked exactly from the frozen data. Changing the support, the seeds, the observable or the model is outside the statement.

Proposition 7.2 (Smooth priors need not concentrate)**.** *For every* $\varepsilon > 0$ *and* $\eta \in (0, 1)$ *there is a smooth, strictly positive, normalized density on the positive part of the window, of the form* $\pi(\lambda) \propto e^{-k\lambda}$*, that gives probability greater than* $1 - \eta$ *to admitted values at which every seed's ratio*

is below ε. One construction works simultaneously for the 32-, 64- and 128-patch populations and throughout the threshold and amplitude ranges of Section 5.3.

This is mechanized for the actual model. A simple numerical instance makes the point: on $[0, 1]$ the density proportional to $e^{-\lambda/0.01}$ is smooth and positive yet gives the interval $[0.1, 1]$ a mass of only about 4.5×10^{-5}. Regularity of a prior near $\lambda = 0$ therefore says nothing about how much mass it puts near the upper edge.

Taken together with Section 6.4, these results say what a typicality argument for the coincidence would need: a prior family specified with quantitative bounds over the whole window, pushed forward through the joint distribution of records and seeds. Neither smoothness, nor the form of the window, nor record counting supplies it. This is the familiar crux of anthropic measure arguments [3], made precise for a model whose window and first-record ratio bounds are proved exactly. The paper does not endorse any prior and draws no anthropic conclusion.

8 What the data say: pre-registered audits

The preceding sections worked in model units. This section turns to public data, and asks only questions that a calibrated audit can answer. The audits test whether fitted corrections to the homogeneous expansion are consistent across declared model families and data splits [30]. Either outcome alone leaves the physical origin of the constant term unresolved: subset dependence can also arise from noise, fitting choices or unmodeled systematics.

The data are DESI DR2 BAO [11], Pantheon+ supernova distances [4] and compressed Planck distance priors, analysed with a compressed likelihood pipeline in which the supernova intercept is profiled analytically. The statistics, splits and thresholds were pre-registered before the audit tables were generated, using data products and baseline fits from an earlier stage of the program (Section 3.4); the amendments are listed there and in Appendix B.

8.1 A rewrite family is constant-compatible

Three models were fitted with the same likelihood and nuisance treatment to $n = 1604$ data points:

model	k	χ^2	ΔAIC	ΔBIC	parameters
ΛCDM	4	1455.80	0	0	$\Omega_m = 0.303$
$w_0 w_a$CDM	6	1442.17	-9.63	$+1.13$	$w_0 = -0.795$, $w_a = -0.673$
rewrite, $E^2 = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + A(1+z)^m$	6	1453.66	$+1.85$	$+12.61$	$A = 0.699$, $m = 0.068$ (best fit)

The $w_0 w_a$ comparison [6, 21] reproduces the familiar survey-era picture: a preference by AIC that BIC does not share [20]. The rewrite family tests whether the correction the data prefer depends on redshift. Its shape parameter is compatible with a constant under the pre-registered criterion $|m| \leq \min(2\sigma_m, 0.25)$, with best fit $m = 0.068$ and $\sigma_m = 0.074$; the posterior median is $m = 0.131$, with 16th and 84th percentiles 0.057 and 0.204. The rewrite also pays an Occam penalty relative to ΛCDM. On these data and this pipeline, then, the fitted correction is consistent with a constant. This says that current geometry does not separate this rewrite family from its constant member; it does not explain why ΛCDM fits, and it does not distinguish a fundamental constant from an effective one.

8.2 One expansion history, two descriptions

A canonical scalar field with a standard two-derivative action and positive energy density cannot reach $w < -1$ without a wrong-sign kinetic term [5, 34], so a preference for crossing $w = -1$ is often read as a problem. The same expansion history can also be written as an additive correction to the matter-plus-radiation expansion rate, $\Delta_H(z) = E^2_{w_0 w_a}(z) - \Omega_m(1+z)^3 - \Omega_r(1+z)^4$ with $E = H/H_0$, which does not by itself specify a microscopic fluid model.

For the CPL form $w(z) = w_0 + w_a z/(1+z)$, the dark-energy contribution to E^2 is

$$\rho(z) = \rho_0 (1+z)^{3(1+w_0+w_a)} \exp\Big(-\frac{3w_a z}{1+z}\Big), \qquad \rho'(z) = \frac{3\,\rho(z)\,(1+w(z))}{1+z}.$$

For $\rho_0 > 0$ and $z > -1$, ρ is positive and smooth, the additive correction Δ_H equals ρ, and, with nonnegative matter and radiation coefficients, $E^2 > 0$. The equation of state reconstructed from the logarithmic derivative of Δ_H is the same $w(z)$, with the same crossing. The density is stationary exactly where $w = -1$, and when $1 + w_0 + w_a \neq 0$ this happens only at $z_\star = -(1+w_0)/(1+w_0+w_a)$. These identities are mechanized. For the best-fit parameters of Section 8.1, $z_\star = 0.4388$; in the additive description the crossing is simply the maximum of a smooth positive density, which ranges from 0.499 to 0.760 over $0 \le z \le 2.5$ (Figure 6a).

The two descriptions are mathematically equivalent ways of writing the same history. A preference for crossing is therefore a statement about the fitted history, and its interpretation depends on which description is taken as physical. The calculation does not show that ghosts are absent from any physical model, and it does not show that dark energy is not a fluid.

8.3 The staging audit flags supernova–BAO subset dependence

Statistic. The rewrite shape is refitted on declared splits of the data (BAO only, supernovae only, and supernovae below and above $z = 0.35$), and the audit records the largest standardized shift from the joint fit,

$$S = \max_{\text{split}} \frac{|m_{\text{split}} - m_{\text{joint}}|}{\sqrt{\sigma^2_{\text{split}} + \sigma^2_{\text{joint}} + 0.05^2}}.$$

The staging fits use a reduced vector of thirteen BAO points and six compressed supernova summaries, with the matter density fixed at a value taken from the earlier fit to the observed data; Planck is not included.

Calibration. Twenty mock ΛCDM realizations through the identical pipeline set the threshold. With twenty mocks and a nominal level of 1/20, the conservative rank rule rejects only above the largest mock score, 8.879. If the observed and mock scores were exchangeable, this rule would bound the false-alarm probability by 1/21; because the mocks are generated at parameters fitted to the observed data, that premise is not established, and the rank fraction is descriptive. (The originally declared interpolated 95th percentile was 8.328.)

Power. The originally declared alternatives applied one rewrite coherently to all probes and were never detected. A common shift of all split estimates cancels in every term of S when the uncertainties are unchanged, so such alternatives are expected to leave the rejection rate at its null value rather than raise it. An audit of staging dependence needs alternatives that break the symmetry the statistic measures. Alternatives applied to one probe at a time do: a supernova distance-modulus tilt $\mu \to \mu + \varepsilon(z/0.5)$ was detected in 17 of 20 mocks at $\varepsilon = 0.20$, and a BAO tilt $\alpha \to \alpha(1 + \varepsilon(z - 0.8))$ in 20 of 20 at $\varepsilon = 0.10$, the smallest tested amplitudes reaching a rate of 0.8 (Figure 6b). With twenty mocks per amplitude these rates are empirical, not population power.

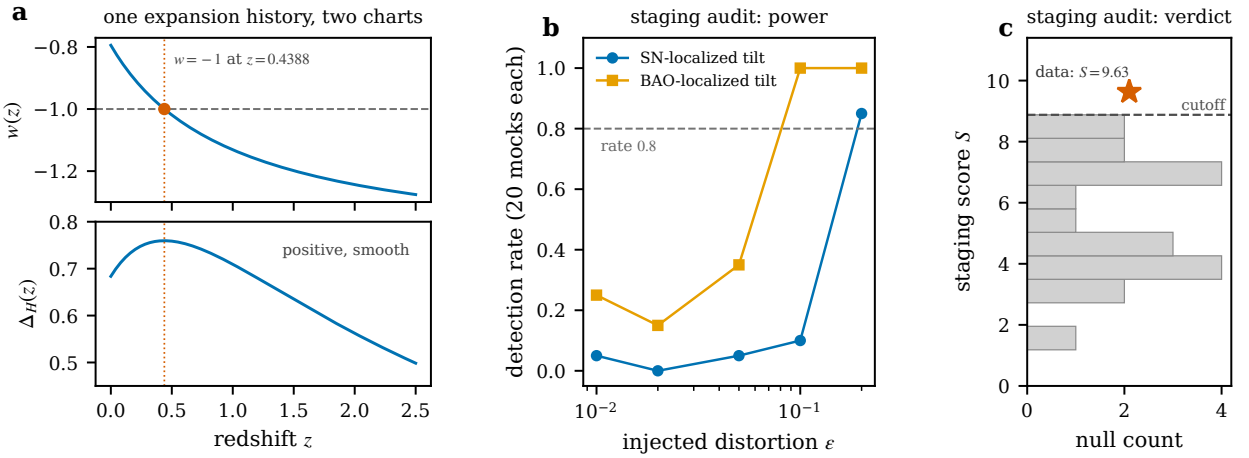


Figure 6. Real-data audits. *(a)* The best-fit w_0w_a history in two descriptions: the fluid equation of state $w(z)$ crosses -1 at $z_\star = 0.4388$ (top), where the additive density $\Delta_H(z)$ has its maximum and stays positive and smooth (bottom). *(b)* Detection rates of the staging audit for probe-localized injections, from 20 mocks per amplitude, at the rank-rule cutoff. *(c)* The 20 null scores, the cutoff (their maximum) and the observed score.

Verdict. On the real data $S = 9.630$, above the cutoff (Figure 6c). The largest shift comes from the high-redshift supernova split, whose fit sits at the upper bound of the amplitude range; the supernova-only and low-redshift supernova fits sit at the lower bound of the shape range. Its size is comparable to the tested supernova tilt at $\varepsilon \approx 0.2$. We read the result as structured probe dependence, consistent with the known mild tension between supernova and BAO distances in this reduced, fixed-anchor pipeline; comparisons of the two kinds of distance are the subject of distance-duality tests [13, 2]. Because the threshold is not a guaranteed level, the result carries no population-level significance, and it does not identify the physical source of the dependence. It is not evidence for a closure mechanism and it is not a statement about Λ.

8.4 Route residuals

The same reduced fits give unsigned residuals between routes: the rewrite amplitude differs between the supernova-only and joint fits by 0.417 and between the BAO-only and joint fits by 0.116; the matter-density residuals are zero because that parameter is held fixed. This demonstrates the protocol on the routes the data allow and inherits the calibration status of Section 8.3. It does not address the H_0 tension, which needs distance-ladder data outside this battery.

8.5 The growth data cannot yet host the decisive test

Structure growth is where a closure correction would be most distinctive. The audit used three public shear and lensing data sets: KiDS-Legacy COSEBIs (126 points, S/N 28.6), HSC-Y3 Fourier-space cosmic shear (170 points, S/N 37.4) and the combined ACT, SPT and Planck CMB-lensing bandpowers (35 points, S/N 61.4). Each data set was cut into shells and each shell classified by a heterogeneity proxy H, the weighted mean excess of the log nonlinear-to-linear power, as slack ($H \leq 0.05$), mild or strong ($H \geq 0.3$). The audit statistic for a shell is the price $\pi = [\lambda_{\max}(\Xi/q^{\mathrm{row}}_{0.95}) - \Omega]_+$ of its adequacy residual, normalized by a null with the same number of

rows and the data covariance, and every null result had to be accompanied by a successful detection of an injected signal. That requirement caught a mis-scaled null normalizer during development.

Three findings follow.

1. **Support.** The slack and strong regimes are measured by different probes. In the computed proxy values, KiDS shells are all mild (H from about 0.088 to 0.172), HSC shells all strong (about 0.477 to 1.911), and CMB lensing has two slack shells and one mild (largest $H \approx 0.144$); full-shape clustering reaches about $H \approx 0.11$ at the scales treated as reliable. None of the surveyed probes spans slack and strong shells within one data set and covariance, so the planned within-family comparison computed no price.
2. **Model dependence.** Where a strong-regime residual can be measured, in HSC, its size depends on the theoretical model: with linear modeling the shell prices are 8.3, 8.8, 16.5 and 10.6, while with nonlinear modeling [28], with or without nuisance parameters, every point price is zero; some confidence intervals keep positive upper limits. These model comparisons use local linearizations around the HSC best fit, propagated to 256 posterior perturbations, rather than full nonlinear evaluations of every posterior draw, and the fitted inputs are not independent data. All stored injection checks pass: the nonlinear configurations detect a finite injected signal (price 3.9), while in the linear configurations the injection adds effectively no signal and so does not demonstrate incremental sensitivity there. The residual is therefore sensitive to omitted nonlinearity; this is not a unique causal attribution.
3. **What a decisive test needs.** One data set with both slack and strong shells under a common covariance and validated nonlinear modeling. Surveys of the LSST, Euclid and Roman class are candidates; whether they meet these support and modeling conditions would still have to be shown.

The CMB-lensing shells, analysed separately as a diagnostic, also give zero point prices with a live injection.

9 The questions revisited

Table 1 collects the results by the questions of Section 1.1. For each it gives what the paper establishes, the evidence type and what remains open.

Read across the rows, the table shows the demarcation the title refers to. The form of the constant slot, as a price of a binding global budget, is a theorem about optimization problems. The window of values that allow records is a theorem about a declared toy universe and its record gate. The value itself is not determined by the readouts of the lattice model, and becomes a gauge coordinate once a global ledger is added. What would fix it, and with what probability, is left to inputs that the models do not contain: a selection rule or a prior with explicit global content. The data audits are consistent with a constant correction on current geometry, flag structured dependence between supernova and BAO subsets, and identify the data a decisive growth test would need.

10 Relation to other work

Unimodular gravity and vacuum-energy sequestering. The allocation model of Section 4 reproduces, in a finite setting, the pattern in which a global constraint with a Lagrange multiplier keeps a uniform energy offset from affecting the state [16, 19]. The closure-price theorem adds a general condition for when such a multiplier is positive, and the gauge theorem of Section 6.4

Table 1. The dark-energy questions and what this paper establishes about each. All model statements are relative to the declared models and readouts.

question	established here	evidence	open
cosmological-constant problem	In the allocation model a common offset is absorbed by the budget price while the budget binds (§4.3); in a finite two-description model any map supporting both readouts must retain the full state (§6.5).	theorem [Lean]	any map from a model budget or readout to gravity and quantum vacuum energy
value of Λ	Convergence readouts do not determine the background offset of the lattice model; with a closed ledger the absolute value is a gauge coordinate that no record-only selector can track and that carries no translation-invariant probability (§6).	theorem [Lean]; the last clause on paper	a selection rule, a physical identification, the value
Hubble tension	Route residuals are reported as a protocol demonstration (§8.4).	audit	calibrator-anchored routes are outside the battery
dataset inconsistencies	A staging audit on supernova and BAO subsets flags structured dependence, led by high-redshift supernovae (§8.3).	audit	a guaranteed false-alarm level; the source of the tension
phantom crossing	The best-fit history crosses $w = -1$ at $z_\star = 0.4388$, where its additive density is maximal, positive and smooth (§8.2).	theorem [Lean] for the identities; fit is numerical	the physical model behind the history
growth or S_8	No single surveyed probe spans both regimes; the HSC residual depends on nonlinear modeling (§8.5).	audit	a single data set spanning both regimes
coincidence	At least ten of twelve seeds keep records exactly for $-1/8000 < \lambda \leq \lambda_+$, $3 < \lambda_+ < 3.4$; the ratio $\lvert\lambda\rvert/\rho_m$ at first record is bounded above but not pinned (§5); no tested prior meets the concentration criterion, and smoothness near zero is insufficient (§7).	theorem [Lean]; computation	a specified prior with global bounds
interacting dark sector, modified gravity	The rewrite family tested is constant-compatible on geometry (§8.1).	audit	splitting observables beyond geometry
anthropic question	Record counting broadens rather than concentrates the declared linear prior (§7.2).	computation	the prior itself

shows that in a closed-ledger model the absolute value of the multiplier is not recoverable from the invariant record. Like unimodular gravity, neither result fixes the value.

Anthropic and measure arguments. The record window has the shape of Weinberg's bound [35]: structure formation limits λ from above without placing the observed value near the limit. The measure probe makes the dependence on the prior explicit [22, 3]. Section 7.3 proves that smoothness near zero does not guarantee concentration, and that reweighting only the frozen base-grid values, with full support and the stipulated conditional seed weights kept, cannot meet the fixed top-decade concentration criterion. The window and its first-record ratio bounds are exact theorems about the model; the prior remains an input.

Backreaction and averaging. A companion pre-registration [30] treats dark energy as a correction that appears when a homogeneous completion fails to commute with inhomogeneous evolution, and the audits of Section 8 are built from its discriminators. The growth result is relevant to any program looking for dynamical signatures in structure data: on the surveyed probes the decisive within-data-set comparison is not available, and an apparent strong-regime residual is sensitive to nonlinear modeling.

Holography. Section 4.4 states the precise finite relation between min-cut area and an optimization price. Both are outputs of a duality, but area is an optimal value under a declared scaling and the closure price is a multiplier; the proposed shared composition rule for entanglement and area fails on an explicit example. We use the relation only as an analogy.

The Six Birds program. Earlier work applied the same calculus to the structure of the Standard Model and to the relation between quantum mechanics and general relativity [32]. That work expected questions about contingent quantities to yield windows and statements about what is undetermined, rather than value laws, and questions about budgets to yield structural relations. The results here fit that expectation: the budget question gave a theorem about prices, and the value question gave readout-relative nonidentifiability and a gauge result. They do not prove it, since the value question was answered for specific readouts and a specific ledger, not for all possible laws. This study also reused the earlier framework deliberately, so it tests the framework's consistency and reach rather than providing an independent confirmation.

11 Limits and open problems

11.1 Limits of the results

The models are small and declared. The allocation model has 32 patches and a sum budget; the record window uses linear growth, a fixed collapse threshold and an imposed erasure rule; the lattice model has eight sites. Their theorems hold for these models. The orbit classification covers the sampled offsets and couplings, not every intermediate value. Several statements are certified by exact computations outside Lean, and the sampled sweeps, measure tables and audit fits are floating-point results without rigorous error bounds. The staging audit is calibrated with twenty mocks at a fitted anchor, so its false-alarm level is not guaranteed, and its detection rates are empirical. Results imported from other Six Birds papers frame the questions but are not proved here.

11.2 What would change the picture

development	what it would settle
a map from a closure budget to a gravitational field equation, with conserved stress-energy and the right dimensions and sign	whether the price of Section 4 can be the physical cosmological constant
a selection rule	a parameter choice for the value question of Section 6, within the models
a physically motivated prior with quantitative bounds over the whole window	a typicality calculation for Section 7, within the models; a prior alone does not fix the realized value
staging or probe dependence of the inferred correction that survives a composite-null calibration and grows with lever arm	it would establish, or its absence would constrain, calibrated inconsistency within the specified likelihood and nuisance model; distinguishing physical origins needs independently specified competing models and predictions
one growth data set spanning slack and strong shells under a common covariance, with validated nonlinear modeling	the decisive within-data-set growth test of Section 8.5, in either direction
a stable fundamental model that produces the fitted crossing history	a stable physical realization of that history; distinguishing among realizations would need further predictions and observations

11.3 Open mathematical problems

Three problems are natural next steps. First, classify the orbits of the lattice model on whole intervals of offset and coupling, including the transition between convergence and two-cycles between $\kappa = 15$ and $\kappa = 20$. Second, mechanize the global existence of the constrained ledger flow of Section 6.4, which is currently a paper-level argument. Third, give the staging audit a calibration with a guaranteed level for a composite null whose nuisance parameters are fitted to the data.

12 Conclusion

We separated the dark-energy problem into questions of different kinds and posed each to a finite model or a data audit. In the models, a strictly sensitive global budget carries one positive price, and, for an originally binding budget, a common linear term shifts that price without moving the allocation as long as the shifted price stays nonnegative; in a toy universe, at least ten of twelve seeds keep records exactly for $-1/8000 < \lambda \leq \lambda_+$ with $3 < \lambda_+ < 3.4$, without pinning the coincidence ratio; the convergence readouts of a self-consistent lattice model do not determine its background offset, and a global ledger turns the absolute value into a gauge coordinate that no record-only selector can track and that carries no translation-invariant probability; and none of five tested priors meets the concentration criterion, while smoothness near zero cannot ensure it. In the data, a rewrite family is constant-compatible, the best-fit evolving history has a regular density through its phantom crossing, a staging audit flags structured dependence between supernova and BAO subsets, and no single surveyed growth probe can yet host the decisive test.

These results make "contingent modulus" a precise, model-relative statement and show what a selection principle or a prior would have to supply. They do not identify the model price with the physical cosmological constant, do not derive its value, and do not refute a fundamental Λ.

The statements labelled as Lean theorems are kernel-checked, and the local numerical results are documented by committed artifacts and reproduction scripts in the accompanying repository.

A Model specifications

All models are finite and deterministic given their seeds, run with single-threaded numerical libraries, and reported in model units. A *validator* is an independently written script that re-derives a verdict from the raw output tables; a nonzero mismatch count fails the run.

A.1 The allocation model (Section 4)

Inputs. $N = 32$ patches indexed $i = 0, \dots, 31$, with fixed heterogeneous coefficients

$$w_i = 1.15 + 0.35\sin(0.37\,i + 0.2) + 0.08\cos(1.13\,i), \qquad c_i = 0.18 + 0.22\,\big(0.55 + 0.45\cos(0.29\,i + 0.4)\big).$$

No random numbers are used. These formulas generate the coefficients in floating point; the saved binary values are reified as exact rational inputs in Lean.

Problem. Maximize $\sum_i [w_i \log(1 + x_i) - \frac{1}{2} c_i x_i^2 + b\,x_i]$ over $x_i \geq 0$ with $\sum_i x_i \leq B$. The optimality conditions are $u_i'(x_i) = p$ for $x_i > 0$, $u_i'(0) \leq p$ for $x_i = 0$, $p \geq 0$ and $p(B - \sum_i x_i) = 0$. The solver finds the price by a bracketed scalar root of the allocation equation, decides activity from the optimizer of the shifted, unconstrained problem, and treats the zero budget separately.

Sweep and readouts. Ten budgets from 8.90 to 66.73; the three largest exceed the unconstrained total and are slack. The price is read from the optimality conditions and, independently, as minus a central finite difference of the optimal cost $V(B) = -\max \sum_i u_i(x_i)$. The largest optimality residual in the recorded sweep is 1.50×10^{-12} (1.11×10^{-12} in a later replay), and the largest price–derivative discrepancy is 3.56×10^{-11}.

Offset leg. The same offset $b\,x_i$ with $b = \pm 0.2$ is applied with the global budget at $B \approx 22.24$ and without any budget. Floating results: with the budget, allocation shift at most 5.7×10^{-14} and price shift b to within 3.3×10^{-14}; without it, allocation shifts of 0.56 and 0.67.

Controls. Per-patch budgets of equal total give unequal prices (0.494, 0.548, 0.568, …); slack budgets give price zero; a closed-form instance is unit-tested. The exact theory (unique optimizer, price monotonicity, C^1 value with $V' = -p$, offset screening on $[-\frac{1}{4}, \frac{1}{4}]$, the original inputs and both controls) is mechanized (Appendix C).

A.2 The record-window model (Section 5)

Background. $H^2 = a^{-3} + \lambda$; integration from $a_0 = 10^{-3}$ to $A = 20$. For $\lambda < 0$ the expansion turns around at $a = |\lambda|^{-1/3}$; turnaround at or before A triggers erasure.

Patches. $N = 64$ per seed; growth $\ddot{D} + 2H\dot{D} = \frac{3}{2} a^{-3} D$ with $D(a_0) = 1$, $dD/da(a_0) = 1/a_0$; amplitudes $m_i = s\,|z_i|$ with z_i standard normal and $s = 10^{-3}$; independent tags with probability 1/2. Unit test: in the matter era the growing mode $D \propto a$ is recovered.

Records and predicate. Patch i collapses when $m_i D(a) \geq \delta_c = 1.686$, freezing its tag. A seed passes if collapsed patches with both tags exist at $a = A$ and no erasure occurred; a value of λ passes if at least 10 of 12 seeds pass. In the exact theory the decimal threshold is $\delta_c = 843/500$ and all 768 realized amplitudes and tags are reified.

Sweep. A signed grid over eight decades in $|\lambda|$, twelve seeds per point, with edges refined by bisection to a factor 2.28. Sampled results: lower edge between -1.18×10^{-4} (inside) and -2.7×10^{-4} (outside), upper edge between 2.33 and 5.31; inside witnesses for $N = 32$, 64 and 128

at 0.83, 2.33 and 2.67. Across positive admitted grid values, the median coincidence ratio over seeds ranges from about 3.5×10^{-6} to 2.1, spanning about 5.8 decades.

Exact results. Window $(-1/8000, \lambda_+]$ with $3 < \lambda_+ < 17/5$; eternal window $[0, u_\infty)$ with $3 < u_\infty < 3.4$; population brackets $(1.3, 1.7)$, $(3, 3.4)$, $(5, 6.25)$; starvation for $\lambda \geq 32$; first-record bounds of Proposition 5.2; regularity in threshold and amplitude scale on $[1.2, 2] \times [\frac{1}{3}, 3]$.

Controls. Zero amplitudes give no collapses; constant tags fail the both-tag seed predicate; the outside witnesses fail for the stated reasons; the validator re-derives every sampled verdict.

A.3 The measure probe (Section 7)

Inputs. The A.2 model, with 36 logarithmically spaced positive values inside the window, eight negative values and zero. Per seed and value: pass flag, N_{rec}, f_{coll}, a_{rec}, $a_\lambda/a_{\mathrm{rec}}$ and R.

Classes and weights. M1 to M5 as in Section 7.1, with cell masses from trapezoid widths in λ or $\log \lambda$. Each cell mass is divided equally among the twelve seeds, each passing seed contributes its own readout with its own record weight, and quantiles are left inverses of the weighted empirical distribution function. Logarithmic classes use the cutoff 10^{-5}.

Criterion. A class concentrates R if $P(\log_{10} R \geq \max - 1) \geq 0.8$ and the central 90% width of $\log_{10} R$ is at most 1.5 decades. Results are in Section 7.2; robustness variants $N = 32$, $N = 128$ and $\delta_c = 2$ give no concentrating class.

A.4 The lattice model (Section 6)

State and dynamics. A complex field $\psi \in \mathbb{C}^8$ on a ring, with kinetic operator $(Ku)_i = u_i - \frac{1}{2}(u_{i-1} + u_{i+1})$ and gradient $(\nabla\psi)_i = \frac{1}{2}(\psi_{i+1} - \psi_{i-1})$. The energy density and potential are $T_{00,i} = |(\nabla\psi)_i|^2 + V_{\mathrm{bg},i}|\psi_i|^2$ and $V_i = V_{\mathrm{bg},i} + \kappa T_{00,i}$. One update computes the normalized lowest eigenvector ϕ of $H = K + \mathrm{diag}(V[\psi])$, replaces it by $\phi e^{-i \arg z}$ with $z = \langle \phi, \psi \rangle$ unless $|z| \leq 10^{-10}$, mixes with weight $1/2$ and normalizes. The initial field and the base background profile, whose values lie in $[0.15, 0.35]$, are those of the source.

Test. After 60 updates, four diagnostics must all be at most 10^{-8}: the last update size; the fixed-point gap $\|\phi' - \psi\|$, where ϕ' is the lowest eigenvector of the final Hamiltonian after the source's realignment, which is applied twice in this diagnostic; the eigen-equation residual $\|H\psi - \langle \psi, H\psi \rangle \psi\|$; and a sourcing residual that, as implemented, compares the potential with itself and is identically zero.

Sweeps. Offsets $c \in \{-8, -5, -3, -2, -1, -0.5, -0.2, -0.1, 0, 0.1, 0.2, 0.5, 1, 2, 3, 5, 8, 10\}$ at $\kappa = 0.3$; couplings $\kappa \in \{0, 0.1, 0.3, 0.5, 1, 2, 3, 5, 10, 15, 20, 30\}$ at $c = 0$. Sixty-update outcomes: offsets -5, -3 and $+10$ fail; couplings 15, 20 and 30 fail. The exact long-run classification is Theorem 6.1. In the exact theory the decimal parameters 0.1 and 0.3 are the rational values of their binary floating-point representations.

Ledger-closed extension. Adds $Q_\kappa = \sum_i (\rho_i + \frac{\kappa}{2}\rho_i^2)$ and its multiplier; see Section 6.4. The allocation form maximizes a strictly concave utility on the simplex $\sum_i x_i = B$ in ledger coordinates $x_i = \rho_i + \frac{\kappa}{2}\rho_i^2$, and its explicit admitted pair has $\kappa = 2$, densities $(1/2, 1)$, ledger amounts $(3/4, 2)$, budget $11/4$, and backgrounds $(0, 1/2)$ and $(1, 3/2)$ with prices $3/2$ and $5/2$.

B Audit details

B.1 Data and likelihood

Planck 2018 compressed distance priors, DESI DR2 compressed BAO and Pantheon+ supernova distances, $n = 1604$ points in the joint fit. The supernova absolute intercept is profiled analytically and the sound-horizon convention is fixed as in the pipeline's validation. The background is computed with a compressed scipy-based pipeline; multi-parameter dark-energy fits use nested sampling (dynesty). No Boltzmann solver is used anywhere in the geometry audits.

B.2 Pre-registration summary

Pre-registered before the audit tables were generated, using existing data products and baseline fits:

- **Models (B1).** ΛCDM ($k = 4$); w_0w_aCDM ($k = 6$); rewrite $E^2(z) = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + A(1+z)^m$ with $A \geq 0$ and bounded m ($k = 6$). Readouts χ^2, AIC, BIC and the posterior of m. Constant-compatibility criterion: $|m| \leq \min(2\sigma_m, 0.25)$, evaluated at the best-fit m.
- **Chart comparison (B1b).** The best-fit w_0w_a history as $w(z)$ and as $\Delta_H(z)$.
- **Staging statistic (B2).** S as in Section 8.3 over the splits BAO only, supernovae only, supernovae with $z < 0.35$ and with $z \geq 0.35$; Planck is excluded because the audit targets supernova and BAO route dependence. Null: mock ΛCDM realizations through the identical pipeline. All splits reported with bound and convergence flags.
- **Route residuals (B3).** Unsigned parameter residuals of the supernova-only and BAO-only fits against the joint fit.
- **Quarantine.** Fitted amplitudes are in-sample parameters, never predictions.

B.3 Model and chart results

model	k	χ^2	AIC	BIC	ΔAIC	ΔBIC
ΛCDM	4	1455.80	1463.80	1485.32	0	0
w_0w_aCDM	6	1442.17	1454.17	1486.45	−9.63	+1.13
rewrite	6	1453.66	1465.66	1497.94	+1.85	+12.61

ΛCDM: $\Omega_m = 0.303$. w_0w_aCDM: $w_0 = -0.795$, $w_a = -0.673$. Rewrite: best fit $A = 0.699$, $m = 0.068$, with $\sigma_m = 0.074$; posterior median $m = 0.131$ (16th and 84th percentiles 0.057, 0.204). Chart comparison: crossing at $z_\star \approx 0.4388$, the rounded analytic root for the stored best-fit decimals (a grid interpolation gave 0.43885); Δ_H between 0.4985 and 0.7596 on the tabulated range, finite and smooth throughout. All 251 tabulated chart rows agree with the analytic density to 10^{-12}.

B.4 Staging calibration and amendments

Original alternatives. A rewrite $\Delta_H = A(1+z)^m$ injected coherently into all probes, at amplitudes up to $A = 5.6$, was never detected. The real score also exceeded the threshold at that stage (9.63 against 8.67), and was not interpreted, because the audit had not been shown to detect anything.

Amended alternatives (logged). A supernova-only tilt $\mu \to \mu + \varepsilon(z/0.5)$ and a BAO-only tilt $\alpha \to \alpha(1 + \varepsilon(z - 0.8))$ with $\varepsilon \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$, 20 realizations each, and the same statistic and null. Detection counts at the rank-rule cutoff 8.879:

ε	0.01	0.02	0.05	0.10	0.20
supernova tilt	1/20	0/20	1/20	2/20	17/20
BAO tilt	5/20	3/20	7/20	20/20	20/20

Threshold rule. The originally declared threshold was the interpolated empirical 95th percentile of the 20 null scores, 8.328. Interpolated percentiles can exceed their nominal false-alarm rate even for exchangeable scores: for a uniformly random permutation of the equally spaced scores $0, \ldots, 20$, with twenty used for calibration and one held out, the interpolated 95th-percentile rule rejects with probability $2/21 > 1/20$. The conservative rank rule rejects only scores above the $\lceil (n+1)(1-\alpha) \rceil$-th smallest null score (never rejecting if that rank exceeds n), here the maximum, 8.879; under exchangeability its rejection probability is at most 1/21. Both the finite counting law and the counterexample are mechanized. With either threshold the real score 9.630 fires, and the detection-grid values remain 0.20 and 0.10.

Split fits.

split	data points	A	m	bound flag	split score
joint	19	0.817	-0.454	none	—
BAO only	13	0.701	0.053	none	6.35
supernovae only	6	0.400	-6.0	m at lower bound	7.10
supernovae, $z < 0.35$	3	0.398	-6.0	m at lower bound	6.33
supernovae, $z \geq 0.35$	3	8.0	1.530	A at upper bound	9.63

The reduced vector consists of thirteen BAO points and six compressed supernova summaries, with Ω_m fixed at 0.3027. Reduced vectors are fragile for shape fits, which is why several fits reach a bound; the 0.05 term in the denominator of S limits the effect of small split variances. An early rewrite fit on this reduced vector ($\chi^2/\mathrm{dof} \approx 6.7$, with bound hits) was not comparable to the baselines; Section 8.1 reports the refit through the full likelihood.

B.5 Route residuals

route pair	parameter	unsigned residual
supernovae only vs joint	Ω_m	0 (fixed)
supernovae only vs joint	A	0.417
BAO only vs joint	Ω_m	0 (fixed)
BAO only vs joint	A	0.116

B.6 Growth audit

Probes: KiDS-Legacy COSEBIs (126 points, S/N 28.61), HSC-Y3 Fourier-space cosmic shear (170 points, S/N 37.39), ACT, SPT and Planck CMB lensing (35 bandpowers, S/N 61.35), with BOSS and DESI full-shape clustering for the mild regime. Heterogeneity proxy $H = \langle [\log(P_{\mathrm{nl}}/P_{\mathrm{lin}})]_+ \rangle_W$, evaluated in one posterior-mean cosmology. Shell census (slack $H \leq 0.05$ / mild / strong $H \geq 0.3$): KiDS 0/3/0, HSC 0/0/4, CMB lensing 2/1/0; full-shape about $H \leq 0.11$ at $k \lesssim 0.2$. The within-data-set comparison computed no price because its support condition failed. In separate HSC diagnostics with local CCL-based linearizations over posterior draws, linear-model shell prices are

8.29, 8.80, 16.50 and 10.57, and nonlinear models with and without nuisance parameters give zero point prices. The nonlinear controls use finite injections and return injected point prices of 3.9. The linear controls already exceed the threshold before injection; their injection search ends at the negligible scale 2^{-80}, so they do not demonstrate incremental sensitivity to an injected signal. The fitted geometry inputs of these diagnostics are not independent out-of-sample data.

C Mechanization and reproducibility

C.1 The Lean development

The theorems are mechanized in Lean 4 (v4.28.0) with a pinned mathlib, in the `lean/` directory of the repository. The main modules, all under `lean/SbtCosmo/`, are:

result	modules
closure price (Theorem 4.1) and variants	`Derivation/ConvexClosurePrice`, `ExtendedClosurePrice`, `NonsmoothConePrice`, `ExtendedNonsmoothConePrice`, `NonsmoothMatrixPrice`, `RelativeSchurStationarity`
adequacy residual (Theorem 4.2)	`Adequacy/Support`, `Adequacy/Reconciliation`, `Derivation/SchurDifferential`, `SchurAdjointStationarity`
allocation model (Section 4.3)	`Derivation/AllocationPrice`, `AllocationSensitivity`, `AllocationScreening`, `AllocationSample`
min-cut comparison and entropy example (Section 4.4)	`Derivation/CapacityScale`, `TensorComposition`
record window (Theorem 5.1) and variants	`Derivation/GrowthPopulation`, `WindowRecords`, `WindowSample`, `WindowErasureWitness`, `WindowEternalRecords`, `WindowPatchCounts`, `WindowPatchCountRatio`, `WindowGlobalStarvation`, `WindowSensitivity`, `GrowthEndpointDerivative`
first-record ratios (Theorem 5.2)	`Derivation/WindowFirstRecord`, `WindowSignedRecord`, `WindowEndpointRatio`, `WindowEternalRatioCertified`
orbit classification (Theorem 6.1)	`Derivation/GroundFixedPoint`, `GroundOriginalContinuation`, `GroundOriginalOffsetSeed`, `GroundOriginalOffsetSixty`, `NativeOriginalSampleResults`

result	modules
readout nonidentifiability (Theorem 6.2)	`Derivation/GroundOriginalAuditNonidentifiability`, `BackgroundSelection`
gauge modulus (Theorem 6.3)	`Derivation/GaugeClosure`
mediator (Theorem 6.4) and commuting completions	`Derivation/AccessFork`, `FieldAccessBlock`
measure bounds (Propositions 7.1, 7.2)	`Derivation/PriorObstruction`, `WindowSmoothPrior`
CPL identities (Section 8.2)	`Derivation/CPLReadout`
rank rule and interpolation counterexample (Section 8.3)	`Derivation/StagingAudit`

Brouwer's fixed-point theorem, used in Theorem 6.1, comes from a ported and transitively checked development pinned in `lean/Gametheory/PROVENANCE.json`. Exact certificates for the orbit classification and the window brackets are generated by scripts in the repository and checked by the Lean kernel; a few auxiliary certificates computed outside Lean are marked as exact computations where they are used.

Verification record. A full registered build of the development completed with 11,775 build jobs. It was followed by 20,798 registered declaration and type queries and a cumulative audit of 32,574 queries for the axioms the proofs use. The only axioms that occur are `propext`, `Classical.choice` and `Quot.sound`; there are no `sorry` placeholders, custom axioms or native-code trust extensions. The receipts are in `results/native_registered_final_verification_2026-10-06/`. The recorded full run of the Python test suite passed 127 tests, before later corrected-helper and deterministic line-ending integrations; a subsequent focused check passed four tests. No full-suite receipt exists for the final Python snapshot. These records are in `results/ground_state_2026-10-02/verification.json`.

C.2 Computations

The original model and geometry-audit computations have build scripts, independent validators and tests; additional computations have their own documented checks:

- **allocation model:** build script `scripts/de_dual_carrier.py`, validator `scripts/de_dual_validate.py`, tests `tests/test_de_dual.py`, outputs `results/DE_dual/`;
- **record window:** `scripts/de_window_carrier.py`, `scripts/de_window_validate.py`, `tests/test_de_window.py`, `results/DE_window/`;
- **measure probe:** `scripts/de_measure_probe.py`, `scripts/de_measure_validate.py`, `tests/test_de_measure.py`; the measure tables of Section 7 are in `results/math_review_2026-10-01/`;
- **lattice model:** the vendored source with `scripts/step54_rederive.py` (an independent classifier), outputs `external/step54_repro/` and `results/DE_step54_repro/`; the floating-point traces of Figure 4 are produced by `paper/contents/figures/orbit_traces.py` from the unchanged source;
- **geometry audits:** `scripts/de_audit_run.py`, `scripts/de_audit_b2_fix.py`, validator `scripts/de_audit_validate.py`, tests `tests/test_de_audit.py`, outputs `results/DE_audit/`; the rank-rule recomputation is in `results/staging_rank_2026-10-02/`;

- **growth audit:** outputs in `results/E5SP/`;
- **figures:** `paper/contents/figures/make_figures.py`.

All sweeps and fits run with single-threaded numerical libraries and committed seeds.

C.3 Provenance of the lattice model

The lattice model originates in another Six Birds repository. Its required source files were copied into `external/step54_repro/` with a manifest recording source paths and SHA-256 hashes, with source commits where recorded (the catalog file from the Foundations series has none); the hashes match, and one file differs from its source only by a disclosed path relocation. The sweep was re-run on this project's hardware, and an independently written classifier, sharing no code with the source, recomputed pass or fail for every offset and coupling from the regenerated residual tables. All rows agree with the source artifacts.

C.4 Recomputing the geometry audits

The geometry audits run without a Boltzmann solver; an optional solver import was made lazy, and the acceptance suite was executed with that import blocked. With the pinned requirements in `results/DE_audit_repro/`:

1. `pytest tests/test_de_audit.py` passes (4 tests);
2. `python scripts/de_audit_validate.py --results-dir results/DE_audit` re-derives the verdicts with no mismatch;
3. `python scripts/de_audit_repro_smoke.py --tolerance-chi2 5.0` performs a reduced optimizer refit of the B1 likelihood and reproduces the stored best-fit χ^2 values within tolerance ($\Delta\chi^2 = -0.008$ for ΛCDM and -0.63 for the rewrite).

These are smoke-level comparisons, not a full replication of the posteriors. A skeptical reader can start with the test suites and validators (seconds), then the model builds (minutes), the Lean build (hours on one machine) and the full audit battery (hours, dominated by nested sampling).

D Review record

External review (June 2026). Before the first release, the package was given to an external reviewer with the frozen repository archive and seven questions: whether a demarcation is a legitimate contribution, whether the models' outputs are built in, whether the real-data firing reflects the instrument or a known systematic and whether the change of injected alternatives amounted to tuning, whether an imported result could carry its grade without local reproduction, whether the treatment of coincidence sharpens or relocates the problem, whether any observed cosmological value entered a model, and what the overall grade should be. According to the review record, the reviewer re-ran the model and measure test suites, asked for imported results to be marked as such and for physical wording to be scoped to the models, and regarded the value question as unresolved by the package. The present version takes the same view of the value question.

Mathematical audit (October 2026). Every claim was then checked against its proof or computation, and the supporting mathematics was rebuilt where needed: the closure-price theorem was mechanized in general form; the record window was proved as an exact continuum statement; the lattice model's orbits were classified exactly; the measure pushforward was recomputed on the joint distribution of parameter values and seeds; the staging threshold was given a conservative rank rule;

and the cross-paper imports were re-examined against their current sources. This audit was carried out by the author's own tooling, combining Lean kernel checking, independent reimplementation and adversarial self-review; it is not an independent human review. Appendix E lists the resulting differences from the first version.

E Version history

Version 1 (17 June 2026; DOI 10.5281/zenodo.20713255). First release.

Version 2 (7 October 2026; DOI 10.5281/zenodo.23218104). Rewritten throughout for readability, with a new abstract and new or redrawn figures, and updated to the mathematical audit of Appendix D. The substantive differences from version 1 are:

1. *Value of the constant.* Version 1 read the lattice model's sixty-update pass/fail table as showing that the background offset has no in-layer value law, that it is "bounded moduli" with no possible selector, and that a coupling control discriminates moduli from constrained parameters. The exact analysis shows that all eighteen offsets converge and that the coupling value 15 converges as well, while 20 and 30 approach two-cycles. Version 2 replaces those claims by the orbit classification, the readout nonidentifiability corollary and the gauge theorem of the ledger-closed extension (Section 6).
2. *Imported comparisons.* The comparison of the value of Λ with the Standard Model's generation count, the record-stability duties concerning proton decay, monopoles and tabletop entanglement tests, the identification of min-cut area with a multiplier, the shared entropy–area composition rule and the noncommutation of the uniform completions are no longer used; the argument that an emergent value cannot be derived by proof is withdrawn. Section 4.4 and Section 6.5 state the precise finite results.
3. *Measures.* The pushforward now weights individual record-bearing seeds; every row of the measure table changed, no class concentrates the coincidence ratio in any variant, and the claim that a prior non-singular near zero would make the coincidence typical is replaced by the counterexample of Section 7.3.
4. *Window.* The sampled two-sided window is replaced by the exact window of Theorem 5.1. The infinite-horizon window is half-open, $[0, u_\infty)$; erasure is an imposed rule at turnaround; and the tag distribution affects the record probability.
5. *Closure price.* The theorem is now mechanized in general form rather than through finite witnesses, and the adequacy residual is treated for singular native probes; the allocation coefficients are described as fixed arrays; and offset screening is stated with the condition that the shifted price stay nonnegative.
6. *Audits.* The staging threshold uses a conservative rank rule (cutoff 8.879 instead of 8.328), with unchanged verdict; the explanation of why coherent alternatives were not detected is narrowed; the supernova-only route residual is 0.417 (version 1: 0.303); the growth shell census and HSC price range are corrected; the phantom-crossing statement is restricted to the existence of two regular descriptions; and the rewrite's BIC is reported.

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